

An associated fracture of the anterior glenoid rim, a bony Bankart lesion (Fig. 24.27d) may be demonstrated on X-ray (Fig. 24.27e) or CT scan (Fig. 24.27f).

A compression fracture of the humeral head posteriorly (Hill-Sachs' lesion) (Fig. 24.27g) or tearing of the posterior or superior labrum may also be present. The history is usually one of acute trauma, either direct or indirect, associated with sudden onset of acute shoulder pain. A patient may describe a feeling of the shoulder 'popping out'. On examination, the dislocated shoulder has a characteristic appearance with a prominent humeral head and a hollow below the acromion. There is a loss of the normal smooth contour compared with the uninjured side. Anterior dislocations of the glenohumeral joint are occasionally associated with damage to the axillary nerve, resulting in impaired sensation on the lateral aspect of the shoulder and deltoid weakness. This should be assessed in any acute dislocation.

Management of anterior dislocation

In a hospital setting, the dislocated shoulder should be X-rayed prior to reduction as a fracture may be present. In most cases this is not practical and the dislocation should be reduced as soon as possible. In these cases, a post-reduction film should be obtained. The sooner the dislocated shoulder is reduced, the easier it usually is to relocate. There are a number of methods to relocate the humeral head onto the glenoid cavity. One method is demonstrated in Figure 24.27h. Injection of 10–15 mL of xylocaine into the joint can reduce pain and muscle spasm and aid reduction.

Once the shoulder has been reduced, it is usually stable unless placed in gross abduction and external rotation. A traditional sling should not be used—when the arm is internally rotated, the Bankart lesion worsens by becoming detached from bone. It is better to position the arm in 30° external rotation (i.e. an external rotation pillow or splint). Non-operative treatment in such a brace for 3 weeks, day and night, reduced the incidence of recurrent dislocation. However, we acknowledge it is awkward for patients to have the arm in external rotation for several weeks and many surgeons do not insist on this rehabilitation method.

Recurrent dislocation and Bankart repair

Shoulder dislocations in young athletes have a high rate of recurrence, leading to chronic shoulder instability. Because of this high incidence of recurrent dislocation, arthroscopy should be considered after shoulder dislocation in the younger athlete. If a Bankart lesion is found at arthroscopy, this should be repaired, either arthroscopically or as an open surgical procedure. Arthroscopic repair of a Bankart lesion can reduce the rate of redislocation to <5%.

Following a Bankart repair, the patient may commence pendular movements with the arm within 24 hours, and maintain the arm in a sling for 3–4 weeks. Once the initial pain from the procedure has subsided, active external rotation movements, to just short of the limit of rotation achieved on the operating table, are commenced. It may be helpful to place the arm in a splint in some abduction and external rotation to limit the amount of anterior capsular shortening. Active rotation exercises and stabilisation exercises can be gradually introduced as pain subsides. By 6 weeks, active strengthening can commence. Return to full sport is often achieved at 3–4 months.

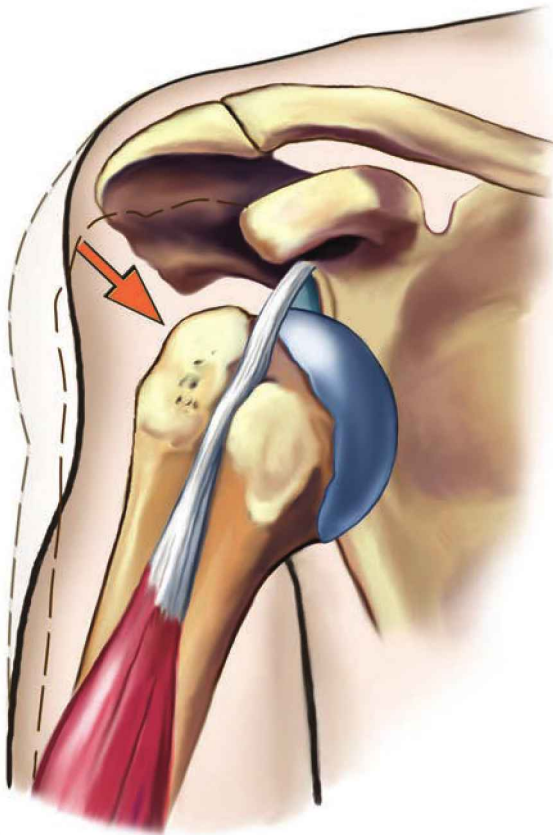
The Latarjet procedure, or a bone block augmentation, is a frequently performed surgical procedure (Fig. 24.28). The choice between a Bankart repair or a Latarjet depends particularly upon the extent of bony damage to the shoulder, the type of sport and the surgeon's preference. The post-operative rehabilitation after a Latarjet procedure is similar to a Bankart repair; however, progression may be slightly quicker. Because the Latarjet procedure involves a bone block procedure it is characterised by faster tissue healing (bone to bone) than the healing of the labrum on the glenoid in the Bankart repair. After an immobilisation period of 3 weeks, gentle mobilisations are allowed, with full recovery of ROM after 8 weeks. Strengthening and stabilisation exercises can start 4–6 weeks after surgery, and return to sports, depending on the specific type of sport, from 12 weeks.

Posterior dislocation of the glenohumeral joint

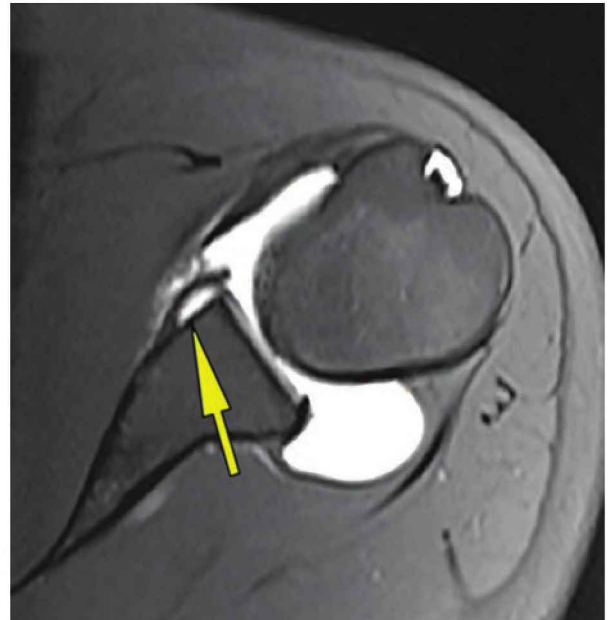
Acute traumatic posterior dislocation is far less common than anterior dislocation. It occurs either as a result of direct trauma or due to a fall on the outstretched arm that is in some degree of internal rotation or adduction. It may also be caused by a seizure of any cause (e.g. electric shock or epileptic fit). The cardinal sign of posterior dislocation is limited external rotation. Suspicion of this diagnosis should be based on the mechanism of injury and the presence of pain and impaired function. Posterior dislocation can easily be overlooked in the AP X-ray. X-ray must include a true lateral or, if possible, axillary view.

Acquired sport-specific instability—AIOS

Overhead athletes very often exhibit acquired glenohumeral instability, characterised by laxity of the anterior capsule, as a result of the extreme use of their shoulder during overhead throwing or smashing activities. This type of acquired instability is often referred to as acquired instability overuse syndrome (AIOS). This kind of 'minor' instability may induce the hyperangulation phenomenon during throwing by excessive anterior translation of the humeral head, causing posterosuperior



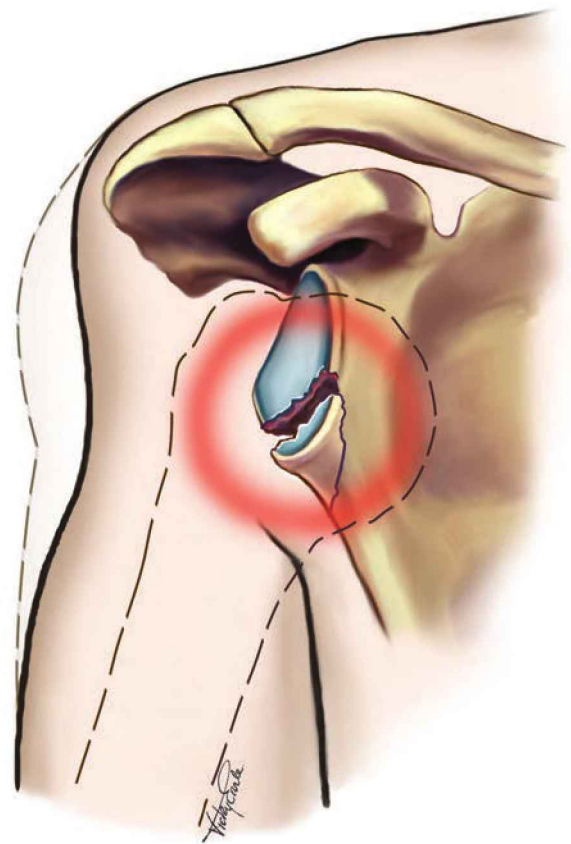
(a)



(c)

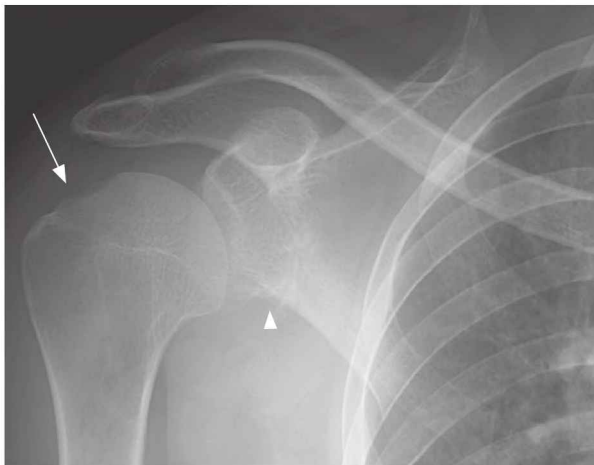


(b)



(d)

Figure 24.27 Dislocation of the shoulder (a) Anterior dislocation of the shoulder disrupts the joint capsule +/- the stabilising ligaments (b) Typical radiographic appearance—the humeral head sits medially over the scapula (c) MR arthrogram axial view showing detachment of the anterior glenoid labrum (yellow arrow), Bankart lesion (d) Bankart lesion showing a fragment of bone separated from the glenoid rim

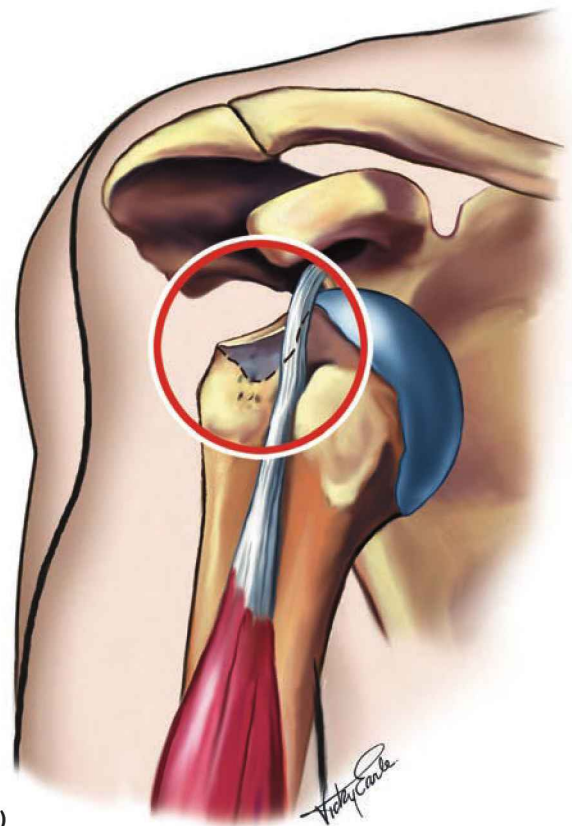


(e)



(f)

Figure 24.27 (cont.) (e) A radiograph showing a Hill-Sachs' lesion (arrow) and bony Bankart lesion (arrowhead) (f) CT scan of a bony Bankart lesion (arrowhead). CT scan may show this pathology where it is undetected by plain radiography (g) Hill-Sachs' lesion showing where the humeral head has impacted on the glenoid rim (h) Position of patient with anterior dislocation of the shoulder to allow reduction of the shoulder. A small weight may be held in the hand to facilitate reduction



(g)



(h)

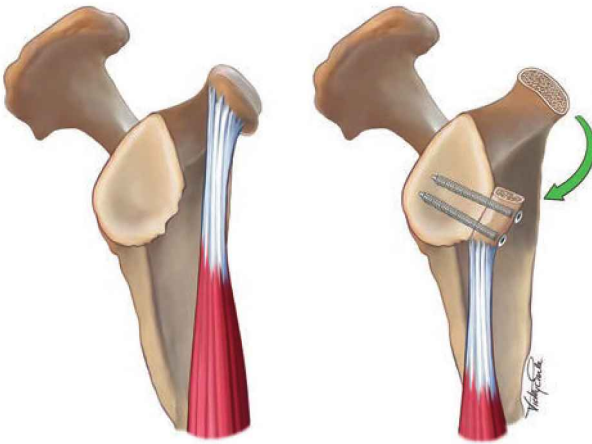


Figure 24.28 Latarjet procedure—one important element of the Latarjet procedure is repositioning the horizontal part of the coracoid to restore bone at the previously damaged anterior inferior margin of the glenoid

impingement. The clinical features comprise recurrent shoulder pain during throwing, possible but not always recurrent subluxations, 'dead arm' syndrome, meaning the athlete reports a sudden inability to throw or smash and a feeling of a 'dead arm'. Chronic pain usually arises from impingement of the rotator cuff tendons, as a result of abnormal humeral head translations, leading to subacromial as well as internal impingement. These recurrent episodes of functional impingement lead to rotator cuff tendinopathy. Apprehension and relocation tests might be painful and/or causing subluxations. These athletes very often show in addition some scapular dyskinesis, GIRD and clinical signs of labral pathology (SLAP lesion).

Atraumatic multidirectional instability—AMBRI

Multidirectional instability of the glenohumeral joint involves a combination of two or three instabilities— anterior, posterior or inferior. Most commonly, multidirectional instability is an atraumatic type of instability, often associated with generalised ligamentous laxity throughout the body. However, it may also result from repetitive trauma, especially at the extremes of motion or, rarely, from a direct blow.

Generalised ligamentous laxity can be assessed by examination of the wrists, thumbs, elbows and knees to determine the presence of hyperextensibility, or obtaining the Beighton score for general laxity.⁶⁷ On examination of the shoulder, besides the functional deficits during active movements and hypermobility during passive

movements, two or three laxity tests will be positive. During the basic clinical examination, one might also find external rotation up to 90° and glenohumeral abduction >90° (Fig. 24.29).⁶⁸

One of the major characteristics of multidirectional instability is that pain in translation occurs in the mid ranges of motion. This indicates a prominent role of altered muscle activation. Lower trapezius and serratus anterior activity is decreased, while pectoralis minor and major and latissimus dorsi increase, creating a position of scapular protraction and glenoid tilting. Frequently the symptoms and pain can be alleviated by placing the scapula in a corrected and stabilised position. Relief of symptoms and decreased translation will point to the need for a therapeutic exercise program for the scapular and shoulder stabilisers. Stretching of the muscles around the shoulder joint should be avoided. If multidirectional instability fails to respond to conservative measures, then surgical treatment may be attempted. However, the results of surgical treatment, particularly in those patients with generalised ligamentous laxity, are not as good as in post-traumatic instability.

A special kind of multidirectional instability is the posterior voluntary subluxation, frequently seen in gymnasts or swimmers with hyperlaxity. Their shoulder 'pumps out' during forward flexion posteriorly, and often the patient is able to perform this subluxation voluntarily. These patients exhibit a marked scapular dyskinesis, characterised by anterior tilting and downward rotation during arm elevation. The posterior subluxation test is positive, and they often exhibit weak external rotation strength.



Figure 24.29 The hyperabduction test

Rehabilitation guidelines for shoulder instability—overview

The rehabilitation guidelines for shoulder instability depend greatly upon the kind of instability the patient exhibits, and the degree or severity of the symptoms. In general, the three kinds of instabilities described need a different rehabilitation approach, described in Table 24.4. The guidelines are aimed to increase local glenohumeral stability as well as to correct aberrant muscle patterning around the shoulder.

The main differences in therapeutic approach are based on the actual best knowledge and best practice regarding muscle patterning and local neuromuscular control in the unstable shoulder, the benefits and the pitfalls of open versus closed chain exercises, and the value of kinetic chain exercises.

Open versus closed chain shoulder exercises

In closed chain exercises, the final link of the chain, in the case of the upper limb, the hand, is supported on a surface, which is fixed or moveable. As a result of the compressive forces in the joint, the patient feels safer to preserve the local glenohumeral stability. In addition, closed chain exercises stimulate normal proprioceptive pathways, enhance local co-contraction in the stabilising muscles, and minimise translations in the midway of motion. However, in view of the functional demands of the upper extremity in daily life, in particular in overhead athletes, requiring functional stability during open chain overhead throwing or smashing, the limitation of the closed chain exercises is that they do not prepare the athlete for full return to sport, and do not load the tissues around the shoulder in the most functional way, thus jeopardising tissue-specific adaptation to training.

In open chain exercises, the hand is free to move in space (with or without an additional resistance) and shear and translational forces are caused in the glenohumeral joint, increasing the challenge for shoulder stability. It is imperative these exercises are implemented in the rehabilitation program of anterior instability (traumatic or acquired); however, only in later stages (for instance after a dislocation), or on the condition that the athlete feels safe performing the exercise (for instance in acquired instability). However, in multidirectional instability, often the only way to initiate and pursue an effective rehabilitation program is to implement mostly closed chain exercises.

Local neuromuscular control of stabilising muscles

Several studies confirm the lack of adequate muscle activity of the local stabilisers. In anterior instability mainly the rotator cuff shows deficiencies in strength, accurate muscle activity and timing of muscle activation. Therefore the first stage in anterior shoulder instability rehabilitation should be to encourage local conscious muscle control in the rotator cuff by teaching the patient how to consciously contract the rotator cuff (with palpation, EMG feedback, local proprioceptive feedback) or by stimulating cuff activity by putting the shoulder in a controlled manner in positions challenging the cuff (Fig. 24.30). However, in AMBRI patients, it is mainly the deltoid, in particular the posterior part, that has recruitment problems. For that reason, deltoid activation should be the main focus in the first stage of the rehabilitation of AMBRI. In addition, since closed chain exercises increase glenohumeral compression, these exercises should be performed in a closed chain (Fig. 24.31). If needed, local taping techniques might facilitate deltoid activation and promote superior translation of the humeral head (Fig. 24.32).

Table 24.4 Rehabilitation guidelines for shoulder instability

TUBS	AIOS	AMBRI
Scapular rehabilitation (below)	Scapular rehabilitation	Scapular rehabilitation
Control of the rotator cuff, followed by rotator cuff strengthening	Control of the rotator cuff, focus on external rotators, followed by external rotator strengthening	Control of the deltoid—posterior cuff
From closed to open chain exercises	From closed to open chain exercises	Mainly closed chain exercises, except if in progression open chain is needed for return to sport
Stretching is not primary goal, but can be added if necessary	Stretching of the posterior shoulder (GIRD) (below)	No stretching
Kinetic chain exercises (below)	Kinetic chain exercises	Kinetic chain exercises



(a)



(b)

Figure 24.30 Exercises stimulating local control of the rotator cuff (a) Prone (b) Supine

Progression in closed chain exercise program—practice

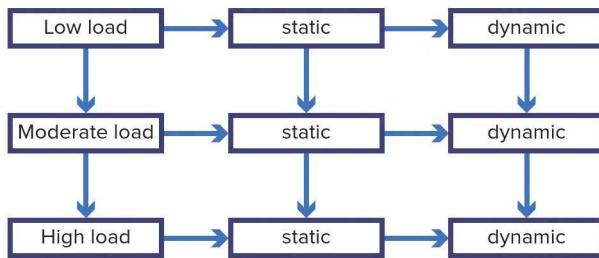
During closed chain exercises, progression is determined by the load (from low or no body weight, over moderate body weight, to full body weight) and whether the exercises are static or dynamic. In static exercises, the shoulder is loaded, but no shoulder movement is required, whereas in dynamic exercises the shoulder moves (from low to high elevation grades). Theoretic progression is shown in Table 24.5 and illustrated in Figure 24.33. It is the therapist's responsibility to find the appropriate progression, based on the specific capacity and instability problems of the patient.



Figure 24.31 Deltoid co-activation in a semi-closed chain



Figure 24.32 Local taping technique for a patient with AMBRI

Table 24.5 Progression in closed chain exercises

(a)



(b)



(c)

Figure 24.33 Progressively more difficult closed chain exercises (a) Low load (b) Moderate load (c) High load**Progression in open chain program—practice**

The open kinetic chain program should commence with basic rotator cuff training (see above); however, it should proceed into more functional exercises, challenging rotator cuff load, functional positions and include eccentric and plyometric exercises. We refer to the thrower's program below for the practical aspect of this progression.

BICEPS-RELATED PATHOLOGY AND SLAP LESIONS

Biceps-related pathologies are a common cause of shoulder pain and disability, and can result in limited work or athletic performance.⁶⁹

Pathomechanics of biceps-related shoulder pain

Pathologic disorders can be divided into three categories: (1) inflammatory/degenerative conditions and partial tears

of the long head of the biceps, (2) instability of the biceps tendon into the bicipital groove, and (3) superior labrum anterior to posterior (SLAP) lesions.⁷⁰ The three categories of disorders may all present with shoulder pain, and although they differ widely in patient populations and pathogenesis, there is significant overlap among the pathologies.

The labrum is the primary attachment site for the shoulder capsule and glenohumeral ligaments. The superior aspect of the glenoid labrum also serves as the attachment site for the tendon of the long head of the biceps muscle (Fig. 24.34). Injuries to the glenoid labrum are divided into SLAP or non-SLAP lesions, and further into stable or unstable lesions. SLAP lesions are injuries to the labrum that extend from anterior to the biceps tendon to posterior to the tendon. They are divided into four types based on the severity and the instability of the biceps tendon (Fig. 24.35).

SLAP lesions are either stable or unstable depending on whether the majority of the superior labrum and the biceps

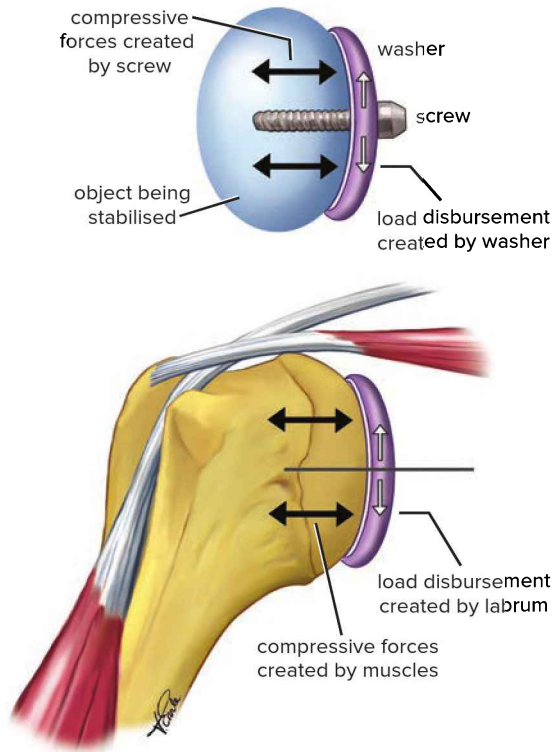


Figure 24.34 The glenoid labrum contributes to proprioception and stability. Dr Ben Kibler compares the stability role with that of a washer (top panel). Like a washer, the deformable labrum (bottom panel), expands the depth and increases the size of the glenoid to increase stability of the interface. The labrum helps to spread load over the glenohumeral joint

tendon are firmly attached to the glenoid margin. Non-SLAP lesions include degenerative, flap and vertical labral tears, as well as unstable lesions such as Bankart lesions.

In general, repetitive overhead activity has been hypothesised as a common mechanism for producing biceps-related shoulder pathology.^{69, 70} Although the pathomechanics are debated, the torsional compressive force on the base of the biceps during the cocking position, as well as the high eccentric activity of the biceps muscle during the follow-through phase of throwing and the impingement of the biceps tendon underneath the acromial arch during overhead activities are believed to possibly cause irritation, dysfunction and failure of the superior labral and biceps tendon complex.⁶⁹

Clinical features

The diagnosis of a glenoid labral tear relies on eliciting a history of an appropriate mechanism of injury, clinical

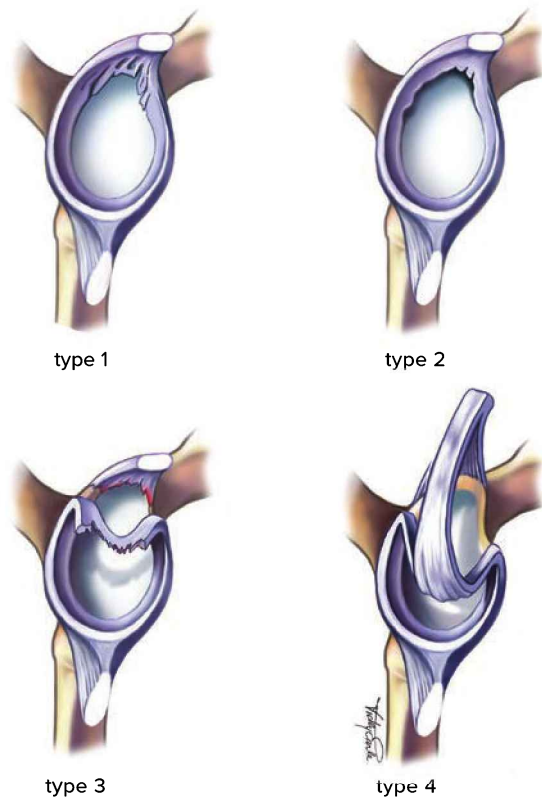


Figure 24.35 Four types of SLAP lesions have been described. In type 1, the attachment of the labrum to the glenoid is intact but there is evidence of fraying and degeneration. Type 2 lesions involve detachment of the superior labrum and tendon of the long head of biceps from the glenoid rim. In type 3 injuries, the meniscoid superior labrum is torn away and displaced into the joint but the tendon and its labral rim attachment are intact. In type 4 lesions, the tear of the superior labrum extends into the tendon, part of which is displaced into the joint along with the superior labrum

assessment and appropriate investigation. The most common mechanisms of injury to the superior glenoid labrum are excessive traction on the labrum through the long head of biceps (e.g. carrying or dropping and catching a heavy object). Throwing injuries occur due to a combination of peel-back traction of the biceps on the labrum in shoulder cocking, abnormal posterosuperior humeral head translation in cocking due to glenohumeral internal rotation deficit, and excessive scapular protraction. Patients complain of pain localised to the posterior or posterosuperior joint line, especially in abduction. Pain in the shoulder is exacerbated by overhead and behind-the-back arm motions. Popping, catching or grinding may also be present.

On examination, there may be tenderness over the anterior aspect of the shoulder and pain on resisted biceps contraction. No current SLAP lesion test is able to guarantee sufficient diagnostic value when used alone, in general an interpretation of 'best-test combinations' is advised in the diagnosis of SLAP lesions (Table 24.2).

Plain radiography is usually unremarkable. MR arthrography, a further refinement of MRI by injection of contrast agent into the shoulder, yields greater detail of intra-articular shoulder structures than does conventional MRI (Fig. 24.36). MR arthrography is particularly useful for the detection and assessment of not only glenoid labral tears, but also small loose bodies, or cartilage flaps. Interpretation of MR arthrograms of the shoulder is best performed by a radiologist with particular expertise in the area, as interpretation is complicated by a wide range of normal anatomical variants. The static MRI is well complemented by dynamic tests which provide a context for interpreting the clinical importance of the MRI.

Treatment of SLAP lesions

In recent clinical guidelines, it is suggested that the vast majority of overhead athletes with shoulder pain should be initially treated with non-operative methods.⁷⁰ Only certain diagnoses, such as traumatic injuries with documented structural damage such as unstable labral

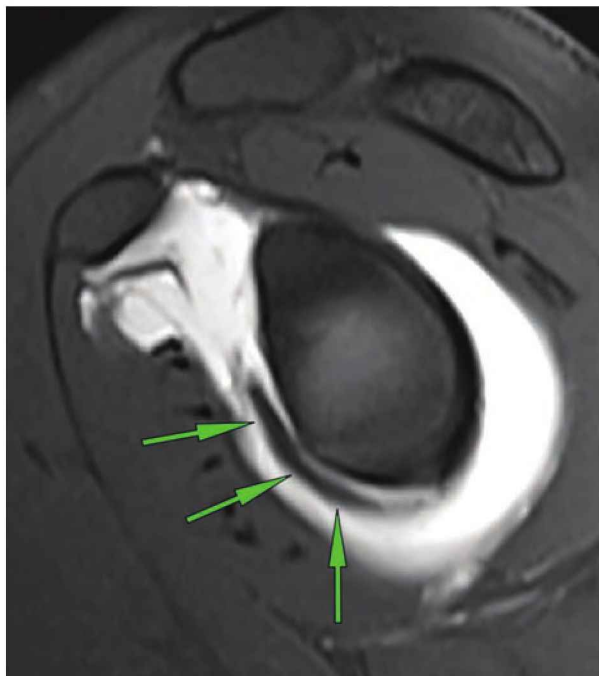


Figure 24.36 MR arthrogram sagittal view showing anterior glenoid labral tear (arrows)

tears, dislocations or rotator cuff tears, may warrant earlier and more aggressive operative intervention.⁷⁰

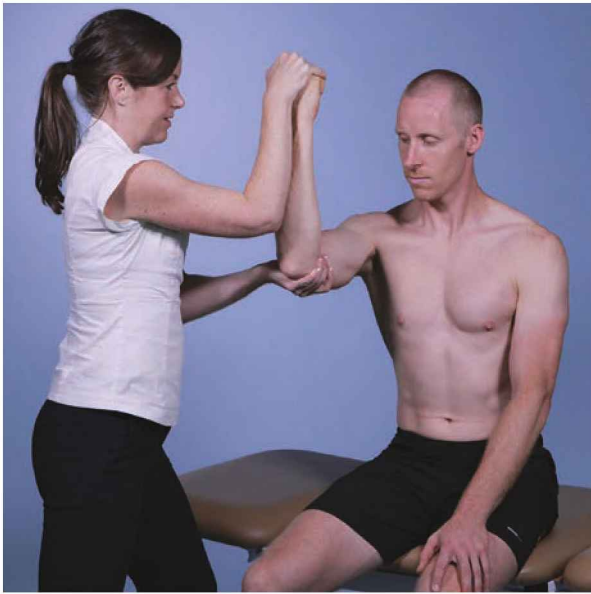
Only a few studies examined the results of conservative treatment in SLAP lesions. Edwards et al.⁷¹ showed that 50% of the conservatively treated athletes returned to play, with similar return to sports as those who had surgery. Fedoriw et al.⁷² concluded from their case series that non-surgical treatment correcting scapular dyskinesia and GIRD had a reasonable success rate in professional baseball players with documented SLAP lesions.

Arthroscopic repair of SLAP lesions results in good to excellent results in patients who are not involved in sports. However, results of surgery in overhead athletes are much less predictable, with a return to previous level of play ranging between 20% and 94%.^{72, 73} In particular, the rate of return to previous performance levels seems to be rather low among specialist throwing athletes such as baseball pitchers.

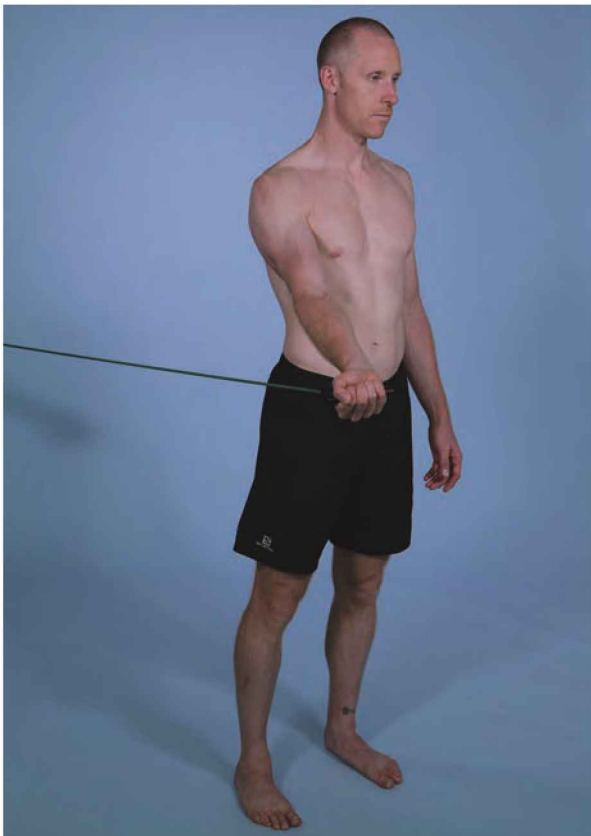
Rehabilitation of biceps-related shoulder pain and SLAP lesions (conservative and postoperative) should follow the general guidelines containing a phased progression of rotator cuff exercises, scapular exercises and stretching. However, tension on the long head of the biceps should be implemented carefully and increased gradually, with early protection of the site of the injury. In addition, in post-operative rehabilitation programs after SLAP repair, biceps activity needs to be controlled during the first 12 weeks following surgery, with no resisted biceps activity during the first 8 weeks to protect the healing of the biceps anchor, and no aggressive strengthening of the biceps for 12 weeks following surgery.⁷⁴

A progressive program consisting of selected rotator cuff and scapular exercises with low to moderate load on the biceps (based on EMG analysis) was proposed, giving the clinician the opportunity to select the appropriate exercises based on the goal and the load on the biceps.⁷⁵ In this continuum of exercises with an increasing level of EMG activity in the biceps, exercises targeting the trapezius resulted in less load on the biceps compared with exercises for the serratus anterior. In addition, exercises with an internal rotation component showed low activity in the biceps, and exercises meant to target the biceps—such as the uppercut exercise (Fig. 24.37a) or resisted forward flexion in supination (Fig. 24.37b)—showed the highest levels of activity in the biceps.

In the advanced stages of rehabilitation, higher load on the biceps is needed to increase its strength in order to prepare the athlete to return to sport, by performing eccentric exercises in throwers or pull-up exercises in gymnasts or pole jumpers.



(a)



(b)

Figure 24.37 (a) The uppercut exercise (b) Forward flexion in external rotation

PATHOLOGICAL GLENOHUMERAL INTERNAL ROTATION DEFICIT

Pathomechanics of GIRD

Posterior shoulder stiffness is the most common adaptation seen at the dominant side of overhead athletes of multiple sports disciplines.⁷⁶ This manifests clinically as decreased glenohumeral cross-body adduction and internal rotation mobility and is believed to be the result of both capsular tightness and muscular contracture. It is hypothesised that the cumulative loads onto the posterior shoulder during the deceleration phase of the throwing motion cause microtrauma and scarring of these soft tissues.

Posterior shoulder stiffness therefore has been suggested to be a causative or perpetuating factor in shoulder impingement and labral pathology.^{3,54,77} Abnormal humeral head translations, caused by selective tightening of the posterior-inferior capsule may decrease the width of the subacromial space, thus causing subacromial impingement. Other studies suggest a posterior and superior translation of the humeral head during cocking with a tight posterior capsule, possibly leading to an encroachment of the rotator cuff tendons against the posterosuperior rim of the glenoid.⁷⁸

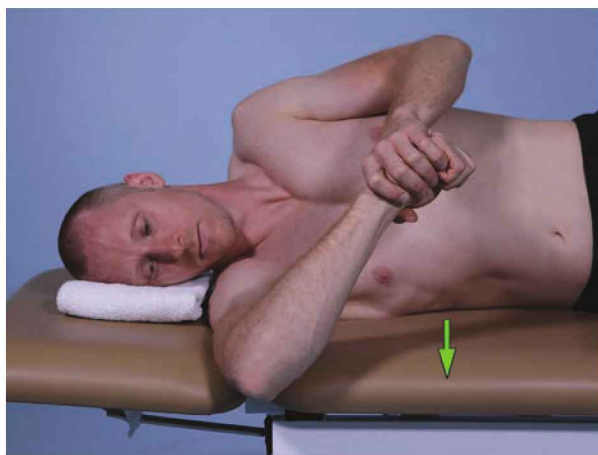
In addition, posterior shoulder tightness seems to affect kinematics of the scapula and the humeral head and is associated with a decreased acromiohumeral distance.⁷⁹ As a result, posterior capsule shortness possibly increases the risk for internal as well as subacromial impingement in the overhead athlete.^{19, 23, 26} A study of elite handball players found that for every 5° increase in total rotational motion, the odds of shoulder injury were reduced by 23%.³

Treatment of GIRD

Given the evidenced impact of posterior shoulder tightness on shoulder kinematics, increasing posterior shoulder flexibility is advisory when mobility deficits exceed the limits associated with increased injury risk. In baseball, reduced internal rotation ROM of 18–25° and reduced total range of motion of 5° are considered to increase the risk for shoulder injury.^{80, 81} Both the cross-body stretch (Fig. 24.38a) and the sleeper stretch (Fig. 24.38b) can be recommended to decrease posterior shoulder tightness.^{81, 82} It was shown that a 6-week daily sleeper stretch program (three reps of 30 seconds) is able to significantly increase the acromiohumeral distance in the dominant shoulder of healthy overhead athletes with GIRD.⁷⁹ Additional joint mobilisation performed by a physiotherapist has a small but non-significant advantage over a home stretching program alone.⁸³ No difference in mobility gain was seen after angular (sleeper stretch and horizontal adduction stretch) and non-angular (Fig. 24.39) joint mobilisation by a physiotherapist.⁸⁴ Muscle energy techniques (hold-relax) during the sleeper stretch and



(a)



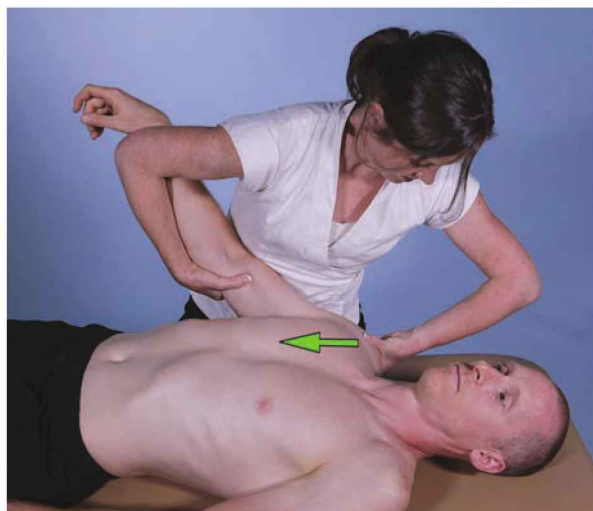
(b)

Figure 24.38 (a) The cross-body stretch (b) The sleeper stretch

the horizontal adduction stretch have proven useful to immediately increase internal rotation range of motion.⁸⁵ Two studies^{84, 86} showed symptom relief after a stretching program in a population of overhead athletes with impingement-related shoulder pain.



(a)



(b)

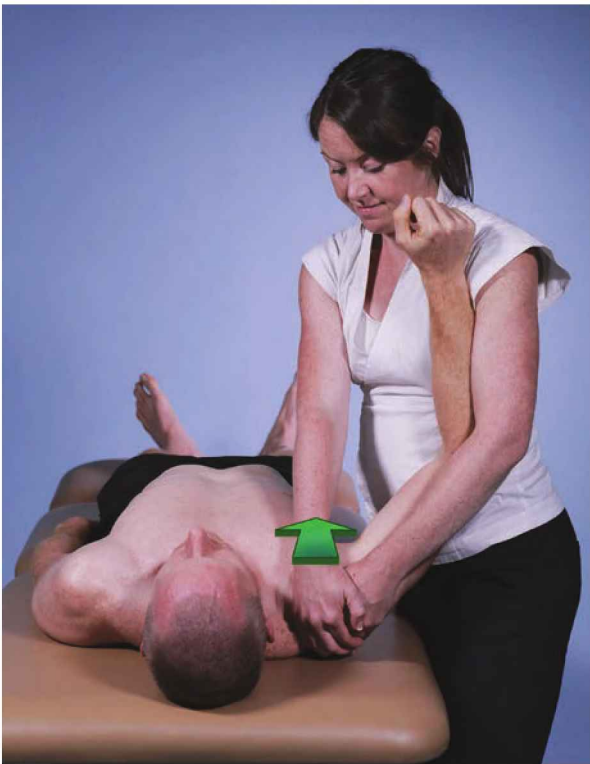
Figure 24.39 (a) Dorsal and (b) Caudal glides

GIRD in the overhead athlete is often associated with acquired instability (AIOS), both leading to imbalances in shoulder kinematics and hyperangulation during cocking. Therefore, in most athletes stretching exercises of the posterior shoulder should be combined with rotator cuff strengthening.

A valuable treatment tool in the mobilisation of the shoulder joint, however, until now, not fully explored or investigated in scientific literature, is mobilisation with movement (MWM). In these techniques, active or passive angular shoulder movements are performed, accompanied by a translational mobilisation of the glenohumeral joint. In view of the arthrokinematic characteristics of impingement, with abnormal anterior and superior translation of the humeral head, in particular dorsal (Fig. 24.40a) and inferior (Fig. 24.40b) glides during active movements are of interest in the treatment of these patients.



(a)



(b)

Figure 24.40 (a) Mobilisation with movement—dorsal glides during forward flexion (b) Inferior glides during scapular elevation

Additional taping to correct humeral head position, as described by McConnell,⁸⁷ might be useful (Fig. 24.41a). However, this should result in immediate symptom relief. In addition, in case of hypertonicity in the infraspinatus, taping with elastic tape (Fig. 24.41b) and dry needling of the infraspinatus (Fig. 24.41c) may relax muscle and

help restore ROM. Home exercises may include using a foam roller (Fig. 24.41d). After resolution of symptoms, the athlete should continue the home stretching program, together with external rotator strength training.

SCAPULAR DYSKINESIS

Most of the studies investigating scapular dyskinesis in relation to shoulder pain have been performed in patients with impingement symptoms and rotator cuff pathology. In general, the identified deviations can be summarised as a lack of scapular upward rotation, posterior tilting and external rotation, increased clavicular elevation and retraction, or characterised by scapular asymmetry in rest or during movement (abnormal scapulohumeral rhythm).

Several authors have demonstrated altered muscle activity patterns in the scapular muscles in patients with shoulder impingement. In particular, researchers consistently demonstrated decreased strength in the serratus anterior, hyperactivity and early activation of

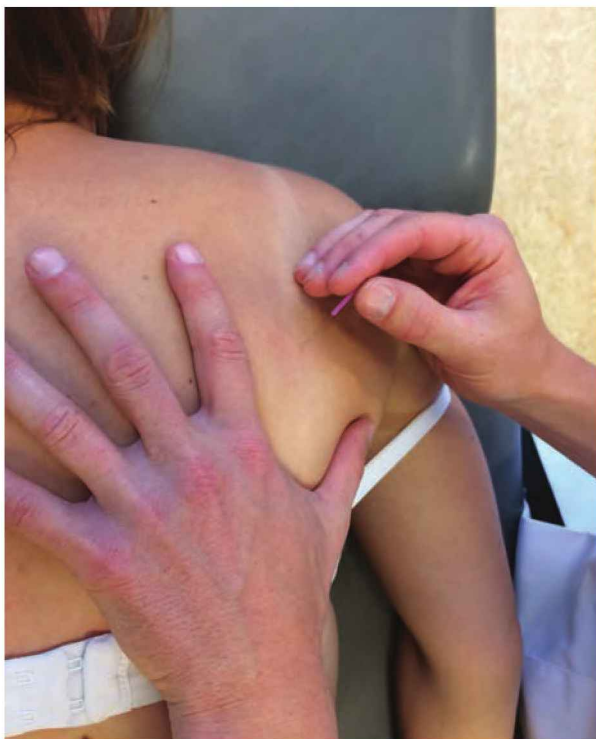


(a)



(b)

Figure 24.41 (a) Shoulder taping to correct humeral head position (b) Elastic tape on the infraspinatus



(c)



(d)

Figure 24.41 (cont.) (c) Dry needling of the infraspinatus
(d) Foam roll exercise for the latissimus dorsi

the upper trapezius (resulting in a shrug—excessive elevation of the shoulder girdle during arm elevation), and decreased activity and late activation of the middle and lower trapezius muscle part. Although some of these studies were performed on non-athletic populations, many of the possible causes for scapular dyskinesis have been established in overhead athletes, and have been suggested to be a contributing and perpetuating factor in the development of impingement symptoms in athletes performing overhead sports.

With respect to soft tissue inflexibility, tightness of the pectoralis minor and posterior glenohumeral capsular stiffness have been established in relation to abnormal scapular position. Increased scapular internal rotation, as

well as increased anterior tilting, has been demonstrated in subjects with a short pectoralis minor or posterior shoulder stiffness. These alterations in scapular position are similar to the scapular deviations established in patients with impingement symptoms, and possibly put the shoulder at more risk for developing shoulder pain.

PRACTICE PEARL

There is no consensus about the cause–consequence relationship between scapular dyskinesis and shoulder pain.

At least six biomechanical mechanisms can potentially alter scapular kinematics. These include pain, soft tissue tightness, muscle activation or strength imbalances, muscle fatigue and thoracic posture. It is unclear whether the alterations found in scapular kinematics are compensatory or contributory to shoulder impingement symptoms and rotator cuff pathology. Regardless of whether the scapular dyskinesis is the cause or the result of painful shoulder conditions, the clinician should focus on scapular evaluation and rehabilitation prior to the treatment of the shoulder. Scapular dyskinesis tends to change the relationship between the glenoid and humerus, to decrease the width of the subacromial space, to jeopardise the fulcrum of the humeral head into the glenoid fossa, and to alter the length–tension relationship of the rotator cuff muscles, hence possibly leading to glenohumeral instability. Moreover, scapular dyskinesis may interrupt the kinetic chain, in particular during overhead throwing and athletic activities.

During overhead movements, the scapula plays a very important role as a link between the trunk and the arm, transferring and increasing the energy and power from the lower extremities and the trunk into the rapidly moving arm. To fulfil this task, the scapula has to be in the correct position and all scapular muscles should be activated at the right time and sufficiently to allow the ground reaction forces to transfer into the shoulder, and finally into the most distal link of the chain, the hand.

Rehabilitation of scapular dyskinesis—a scapular rehabilitation algorithm

The algorithm presented in Figure 24.42 guides the progression of exercises for rehabilitation of scapular dyskinesis.⁸⁸ The upper part of the algorithm summarises the possible causes of scapular dyskinesis, divided in flexibility and muscle performance deficits. The lower part of the algorithm offers guidelines for progressive rehabilitation. Since scapular dyskinesis nearly always is caused by both kind of deficits, the rehabilitation program

Dr Ann Cools' Scapular Rehabilitation Algorithm

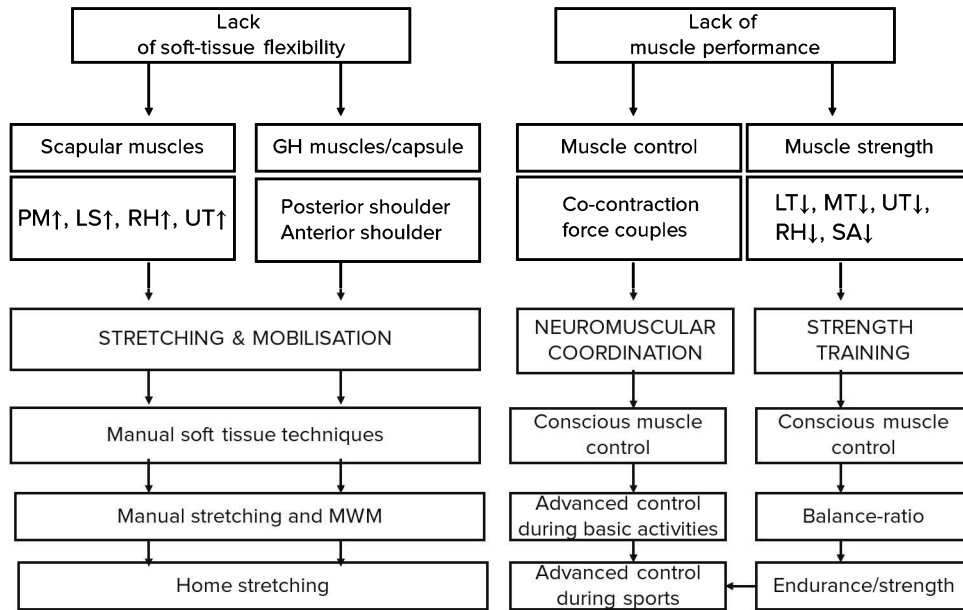


Figure 24.42 Scapular rehabilitation algorithm

PM = pectoralis minor, LS = levator scapulae, RH = rhomboid, UT = upper trapezius, MT = middle trapezius, LT = lower trapezius, SA = serratus anterior, GH = glenohumeral

COOLS ET AL.⁸⁸ REPRODUCED WITH PERMISSION FROM *BRITISH JOURNAL OF SPORTS MEDICINE*

is multifactorial, including stretching as well as muscle recruitment and strengthening exercises.

Rehabilitation of flexibility deficits

Several stretching techniques have been described to increase pectoralis minor length. Superior effects of the 'unilateral corner stretch' (performing passive horizontal abduction with the shoulder in 90° of abduction and external rotation) over 'sitting manual stretching' (in which the therapist performs scapular retraction with the shoulder in a neutral position) and 'supine manual stretch' (similar to the unilateral corner stretch, but performed by the therapist with the patient in a supine position) were established in a study on healthy subjects.⁸⁹ However, from a clinical perspective, these stretches (with the exception of the 'sitting manual stretching') put the athlete's shoulder into a position, possibly causing pain in the case of subacromial or internal impingement.

Therefore in clinical practice, the pectoralis minor might be stretched by performing passive retraction and posterior tilting of the scapula with the shoulder in a neutral or small elevation position and slight external rotation. In particular, direct pressure on the coracoid process provides an intense stretching effect on the

pectoralis minor (Fig. 24.43a). In healthy athletes, the corner stretch might be advised, taking care not to stretch the anterior shoulder capsule too much. An alternative self-stretch exercise, resulting in scapular retraction and posterior tilt is given in Figure 24.43b.

With respect to stretching techniques of the posterior capsule, we refer to the previous section.

Rehabilitation of muscle performance deficits

In view of the new insights and research findings on the role of the scapula in shoulder pathology, current exercise protocols emphasise scapular muscle training as an essential component of shoulder rehabilitation. In the early stage of scapular training, conscious muscle control of the scapular muscles may be necessary to improve proprioception and to normalise scapular resting position. To selectively activate the scapular stabilisers, the 'scapular orientation exercise' has been described.⁹⁰ Prior to high-intensity strength training, the clinician should focus on facilitating correct timing and control of muscle recruitment. Strength training in the presence of faulty scapular control probably might reinforce poor kinematics and fail to reduce pain or improve function.



(a)



(b)

Figure 24.43 (a) Direct manual stretching of the pectoralis minor—therapist performs a posterior tilt of the scapula with direct pressure on the coracoid process (b) Home stretching for the pectoralis minor

Depending on the results of the clinical examination, the therapist may decide in the second stage of scapular muscle training to focus more on muscle control (appropriate co-activation of the scapular force couples) or muscle strength (in case of isolated strength deficit in one or more scapular muscles).

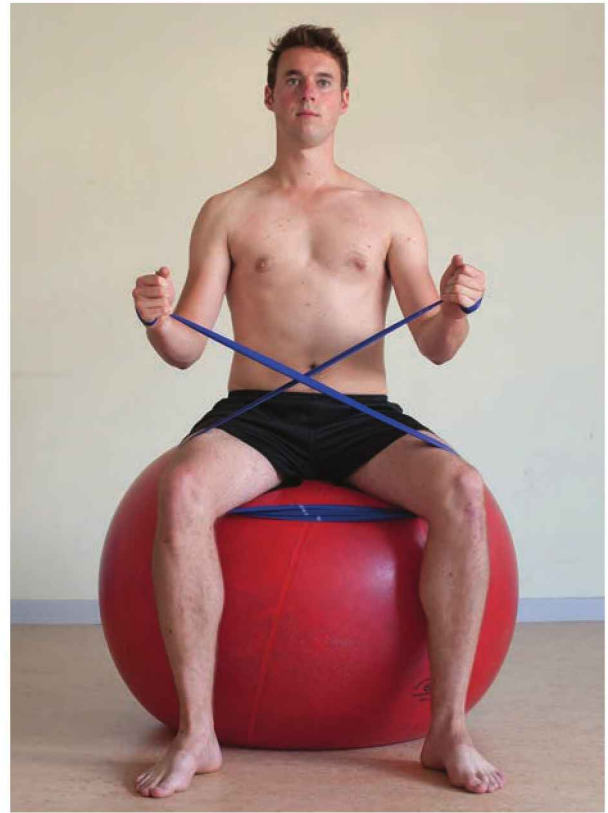
Following the scapular orientation exercises, scapular co-contraction may be trained in basic positions, movements and exercises, activating the key scapular-stabilising muscles without putting high demands on the shoulder joint. In this stage of scapular rehabilitation, very often closed chain exercises are used. They are believed to improve dynamic glenohumeral stability through stimulation of the intra- and periarticular proprioceptors, and enhance co-contraction of the rotator cuff, thus being beneficial in case of shoulder instability. However, open chain functional exercises are also imperative in this stage. Exercises with an external rotation component tend to improve proper scapular muscle recruitment (Fig. 24.44).

For patients with a strength deficit and muscle imbalance in the scapular muscles, selective activation of the weaker muscle parts with minimal activity of the hyperactive muscles is an important component in the second stage of scapular muscle rehabilitation. Because of relative inactivity in lower trapezius (LT) and serratus anterior (SA), often combined with excessive use of upper trapezius (UT), exercises with a low UT/LT, UT/MT and UT/SA ratio are particularly important (Figs 24.45, 24.46).⁹¹ Several studies have examined the effectiveness of a scapula-based rehabilitation program.^{60, 92, 93} Although there are those who advocate for the primacy of treating scapular mechanics, there is no evidence to suggest this has clinical benefits over a comprehensive shoulder treatment program as outlined in this chapter.

In the third stage of scapular rehabilitation, in which the treatment goal is to exercise advanced scapular muscle control and strength during sport-specific movements,



(a)



(b)

Figure 24.44 Examples of exercises with glenohumeral external rotation component (a) During elevation, (b) Combined with core stability and hip abductor training

special attention is given to integration of the kinetic chain into the exercise program, implementation of sport-specific demands by performing plyometric exercises and eccentric exercises. Scapular control should be automatic and integrated into all sport-specific exercises. Throwing athletes should perform exercises in which the external rotators are eccentrically loaded (see thrower's program in Chapter 8).

On the other hand, swimmers should not focus on plyometrics, but rather train in sport-specific positions such as lying prone or supine in situations demanding a lot of core stability and demanding a large number of repetitions. Gymnasts and athletes performing climbing sports should perform high-level closed chain exercises, such as side- and prone-bridging, and sling exercises might



(a)



(b)

Figure 24.45 Four exercises to improve UT/MT and UT/LT ratio (a) Side-lying external rotation, (b) Side-lying forward flexion



(c)



(d)

Figure 24.45 (cont.) (c) Prone horizontal abduction with external rotation (d) Prone extension



(a)



(b)

Figure 24.46 Two exercises to improve UT/SA ratio
(a) Elbow push-up, (b) Serratus push-in closed chain

be added to their program. These exercises have been shown to highly activate the large glenohumeral muscles compared to exercises on a stable surface.⁹²

Throughout the rehabilitation program, and given the evidence that scapular dyskinesis is often related to thoracic kyphosis or flexed thoracic postures, thoracic posture should also be addressed in the rehabilitation of patients with shoulder pain and related scapular dyskinesis. This includes correction of posture into extension, and also mobilisation exercises into extension (Fig. 24.47a) and rotation (Fig. 24.47b), and core-stability training and muscle training of the trunk muscles. The use of therapeutic tape (Fig. 24.48) can be helpful in this stage of the rehabilitation, since it has been shown that taping the thoracic spine and the scapula into extension, posterior tilting and retraction improves trunk posture and shoulder range of motion.

FRACTURES OF THE CLAVICLE

Clavicle fractures are usually caused by either a fall onto the point of the shoulder, for example in horse riding or cycling, or by direct contact with opponents in sports such as football.

The clavicle usually fractures in its middle third with the outer fragment displacing inferiorly and the medial fragment superiorly. It is extremely painful. On examination, there is localised tenderness and swelling and the bony deformity may be palpated. With clavicular shortening or angulation, the scapula will assume a protracted position.

Radiography reveals the fracture. There has been a change in perspective about imaging clavicle fractures as it is sometimes difficult to assess true clavicle position using radiographs. Thus, overlap and shortening need to be carefully monitored for the first 3–4 weeks of non-operative management because significant deformity can occur.

The principles of treatment are to provide pain relief. Clavicle fractures almost always heal in 4–6 weeks. However, often the ends overlap and the clavicle is foreshortened. A foreshortened clavicle is associated with significant functional deficits. A figure-of-eight bandage is designed to prevent foreshortening and has significant advantages over a sling or collar and cuff. During this time the patient should perform self-assisted shoulder flexion to a maximum of 90° to prevent stiffness of the glenohumeral joint.

These fractures⁹⁴ are best managed conservatively and usually heal well. Early surgical fixation is indicated if there is compromise of the skin by bony fragments or foreshortening of greater than 1–2 cm. Occasionally, non-union of a fracture of the clavicle may occur with a fibrous pseudoarthrosis forming. This is treated surgically by open reduction and internal fixation with a dynamic



(a)



(b-i)



(b-ii)

Figure 24.47 (a) and (b-i, ii) Exercises promoting trunk rotation

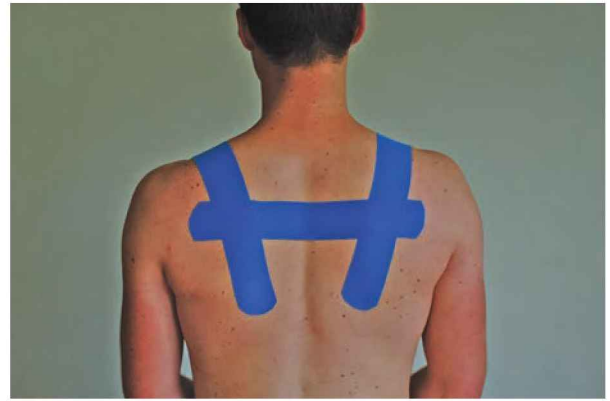


Figure 24.48 Taping technique to promote scapular posterior tilt and retraction

compression plate and bone chips to ensure the length of the clavicle is maintained.⁹⁴

Distal clavicle fractures comprise 12-15% of all clavicle fractures. Many of these fractures involve disruption of the AC joint and/or coracoclavicular ligaments. These fractures are more prone to non-union and delayed union. Classification for these fractures is shown in Table 24.6

Generally, fractures medial to the ligament attachments have greater displacement of fracture fragments and this is associated with increased risk of delayed or non-union if treated non-operatively.⁹⁴

Minimally displaced fractures distal to the coracoclavicular ligament attachments (type I) may be treated with a sling for comfort and early range of motion and isometric strengthening exercises. If displacement is present, then rehabilitation should progress slowly, with active range of motion exercises only introduced when pain resolves and healing has begun radiographically.

Treatment of the more medial (type II) fractures is more controversial.⁹⁴ As there is a high rate of non-union, surgical treatment is often recommended. Distal intra-articular fractures (type III), if stable, should be treated non-surgically as they tend to heal with minimal dysfunction.

Table 24.6 The American Shoulder and Elbow Society classification of distal clavicle fractures

Type	Pathology
I	Fracture distal to coracoclavicular ligaments with little displacement
IIa	Fracture medial to coracoclavicular ligaments
IIb	Fracture between coracoclavicular ligaments
III	Intra-articular fracture without ligament disruption

Special note—immature skeleton

The treatment of distal clavicle fractures in the immature adult is different from that in the adult. Even fractures that present with significant displacement are stable and will eventually heal in an anatomical position. This is due to the fact that, although the fracture is medial to the coracoclavicular ligament attachment, the periosteal envelope remains attached to the coracoclavicular ligaments. The haematoma and subsequent new bone formation stimulated by the periosteum results in remodelling and complete union.

ACROMIOCLAVICULAR JOINT INJURIES

The AC joint is another common site of injury in athletes who fall onto the point of the shoulder. Stability of the AC joint is provided by a number of structures. In order of increasing importance, these are the joint capsule, the AC ligaments and the coracoclavicular ligament comprising the conoid and trapezoid ligaments (Fig. 24.49).

Acute acromioclavicular joint injuries

A modified classification system by Rockwood describes six different types of AC joint injuries (Fig. 24.50). Type I injury corresponds to sprain of the capsule of the joint and is characterised clinically by localised tenderness and pain on movement, especially horizontal flexion. Type II injuries correspond to a complete tear of the AC ligaments with sprain of the coracoclavicular ligaments. On examination, as well as localised tenderness, there is

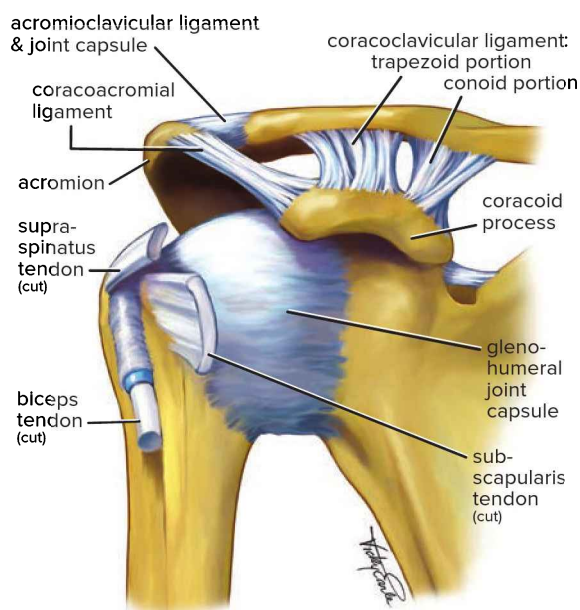


Figure 24.49 The acromioclavicular joint

a small palpable step deformity. Type III and V injuries consist of complete tears of the coracoclavicular ligaments, the conoid and trapezoid. In type III and V injuries, a marked step deformity is present (Fig. 24.51).

Type V injuries can be distinguished from type III injuries radiographically by the amount of displacement. A type V injury has between three and five times greater coracoclavicular space than normal, whereas a type III injury has 25–100% greater coracoclavicular distance than the uninjured side. Type V injury typically involves much greater soft tissue injury and includes damage to the muscle fascia and occasionally the skin. Type IV injuries are characterised by posterior displacement of the clavicle and type VI injuries have an inferiorly displaced clavicle into either a subacromial or subcoracoid position. Type IV, V and VI injuries also have complete rupture of all the ligament complexes and are much rarer injuries than types I, II and III.

Management is based on the general principles of management of ligamentous injuries. Initially, ice is applied to minimise the degree of damage and the injured part is immobilised in a sling for pain relief. This may be for 2–3 days in the case of type I injuries or up to 6 weeks in severe type II or type III injuries. Isometric strengthening exercises should be commenced once pain permits. Return to sport is possible when there is no further localised tenderness and full range of pain-free movement has been regained. The athlete may feel more comfortable on return to sport by taping the AC joint (Fig. 24.52).

The major functional problems in a high grade (III–IV) injury result from loss of strut function to stabilise the scapula, glenohumeral joint and arm. Most type III AC separations showed an alteration of scapular mechanics.^{95,96} Thus, much more consideration should be given to reconstruction in those patients exhibiting altered scapular mechanics.

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The treatment of type III injuries is controversial. Historically, most of these injuries have been treated surgically. However, most clinicians now consider conservative management to be equally effective.

Surgery is clearly indicated for type IV, V and VI injuries and those type III injuries that fail to respond adequately to conservative management.⁹⁷

Chronic acromioclavicular joint pain

Chronic AC joint pain may occur as a result of repeated minor injuries to the AC joint or following a type II or

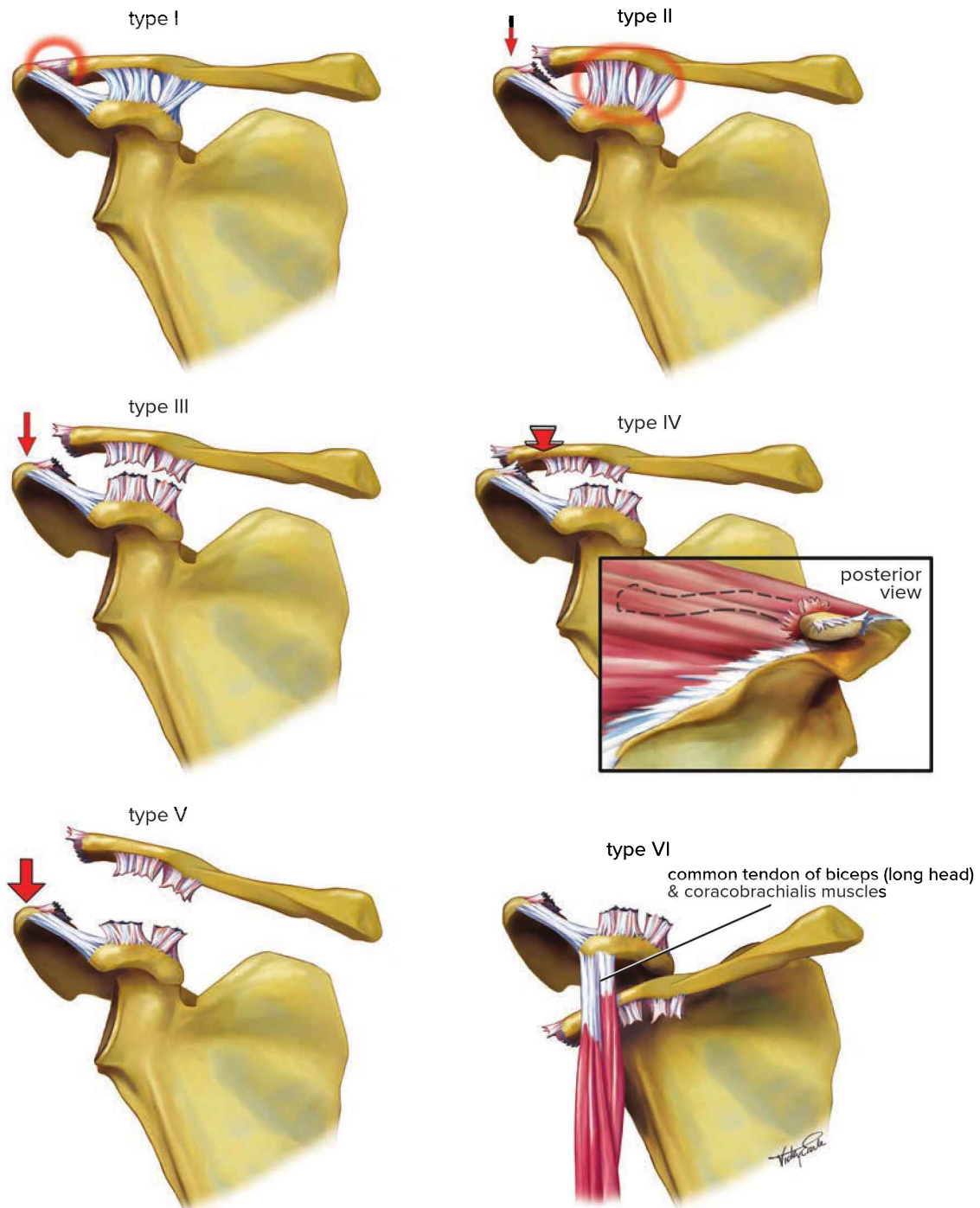


Figure 24.50 Classification of acromioclavicular joint injuries

type III injury. This can damage the fibrocartilaginous meniscus situated within the AC joint.

Another cause of chronic AC joint pain, osteolysis of the outer end of the clavicle, is seen occasionally, especially in weightlifters performing large numbers of

bench presses. X-ray in this condition shows a 'moth-eaten' appearance of the distal end of the clavicle (Fig. 24.53). Horizontal flexion is painful. Rotator cuff impingement may occur due to the abnormal scapular position that results from loss of the clavicle strut. Treatment consists



Figure 24.51 Marked step deformity at the acromioclavicular joint in type III injury



Figure 24.52 Taping after acromioclavicular joint injury

of physiotherapy, including pain relieving modalities and mobilisation, combined with muscle strengthening.

Osteoarthritis of the AC joint may occur as a result of recurrent injuries. This is characterised by a typical radiographic appearance with sclerosis and osteophyte formation (Fig. 24.54).

AC joint pain is usually localised over the joint. Symptoms may be reproduced by AC joint compression using O'Brien's test (with both test positions giving symptoms) or cross-arm adduction. An injection of local anaesthetic and corticosteroid into the AC joint can confirm the diagnosis and provide effective symptom relief.

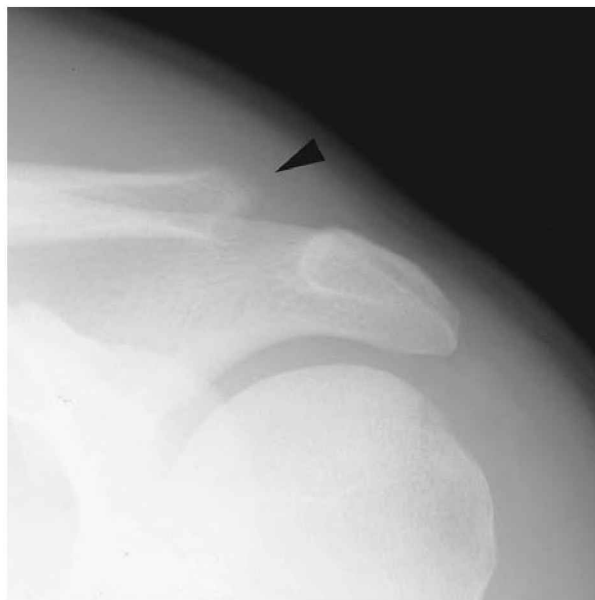


Figure 24.53 Osteolysis of the outer end of the clavicle, showing a 'moth-eaten' appearance



Figure 24.54 Post-traumatic osteoarthritis of the acromioclavicular joint showing a spur and bony irregularity on the acromion

Persistent AC joint pain may require arthroscopic distal clavicle excision.⁹⁸ The surgeon must take care not to disrupt the ligamentous attachments to the distal clavicle. More than 5 mm resection alters the joint loads and removes the bony attachments.⁹⁸

LESS COMMON CAUSES OF SHOULDER PAIN

As shown in Table 24.1, there are a number of less common causes of shoulder pain. These include other muscle tears, adhesive capsulitis, neurovascular injuries, other fractures and the 'snapping scapula'.

Other muscle tears around the shoulder

Other muscle tears around the shoulder include the long head of biceps, pectoralis major and subscapularis.

Rupture of the long head of the biceps

Rupture of the loaded long head of the biceps muscle may occur in the older athlete and is usually accompanied by immediate sharp pain and a tearing sensation.

The deformity is obvious—the muscle is detached from its proximal attachment and bunches up in the distal arm. The deformity is accentuated by contraction of the biceps. Often there is little ongoing pain. Surprisingly, biceps strength is almost fully maintained. Imaging is via MRI or ultrasound. Those who do not rely on their upper arm in sport are generally satisfied with reassurance and require no definitive treatment. In those who perform power sports, surgery may be indicated.

Pectoralis major tears

Pectoralis major tears may be partial ruptures (grades I-II) or complete (grade III).⁹⁹ Complete rupture occurs at the site of its insertion to the humerus. This is usually seen in weight training, especially when performing a bench press. The typical history is of sudden onset of pain on the medial aspect of the upper arm. Examination reveals localised tenderness and swelling. Resisted contraction of the pectoralis major is weak and may be painful.

A partial tear is treated conservatively with ice and a strengthening program over a period of 4–6 weeks. A complete tear of the pectoralis major should be treated by surgical repair.¹⁰⁰ It is usually possible to differentiate between a partial and a complete tear clinically. Ultrasound or MRI examination may assist in this differentiation.

Subscapularis muscle tears

Tears of the subscapularis muscle can occur with sudden forceful external rotation or extension applied to the abducted arm. There is usually no associated instability. The main complaint is pain, and range of motion may be maintained or even increased. On examination, the patient will have increased passive external rotation with the shoulder adducted at the side, weakness of internal rotation and a positive lift-off sign (Fig. 21.2h). MRI and ultrasound will confirm the diagnosis. Treatment should be immediate surgical repair.

Adhesive capsulitis—'frozen shoulder'

Glenohumeral joint stiffness is not uncommon after significant trauma, for instance, a fracture or surgery. It may also follow an injury to the neural structures in the neck or it may occur spontaneously.

The age group in which spontaneous shoulder stiffness, commonly referred to as adhesive capsulitis, occurs is between 40 and 60 years of age. Idiopathic adhesive capsulitis more commonly affects the non-dominant shoulder and is more prevalent in women than men (1.3:1). Adhesive capsulitis is more common in patients with diabetes and there is an association with thyroid disorders and drugs that involve inhibition of matrix degradation.

The diagnosis of shoulder stiffness is relatively easy to make by evaluating passive external rotation with the elbow at the side. Care should be taken to stabilise the scapula when examining for shoulder stiffness as significant range of motion can occur at the scapulothoracic articulation. Surgical or post-traumatic scapular stiffness usually resolves within 12 months and further surgical intervention is rarely necessary. Physiotherapy may be valuable.

Idiopathic adhesive capsulitis is a self-limiting condition that resolves, on average, over 1.5 years.¹⁰¹ There is no evidence that intensive physiotherapy, injections¹⁰² or drugs change the outcome of idiopathic adhesive capsulitis. However, pain-relieving techniques and modalities, as well as patient education during the painful phase are useful and appreciated by the patient. During the resolving phase, graded mobilisation and a home stretching and strengthening program assist the patient in the natural recovery process.

Arthroscopic capsular release to divide the thickened shoulder capsule and an early aggressive supervised range of motion program are not effective at restoring motion and relieving pain in patients with idiopathic adhesive capsulitis.^{103, 104}

Neurovascular injuries

Suprascapular nerve entrapment

The most common entrapment is of the suprascapular nerve.¹⁰⁵ The suprascapular nerve is derived from the upper trunk of the brachial plexus formed by the roots of the C5 and C6 nerves. The course of the nerve is shown in Figure 24.55. The nerve passes downwards beneath the trapezius to the superior border of the scapula. Here it passes through the suprascapular notch. The roof of this notch is formed by the transverse scapular ligament. After passing through the notch, the nerve supplies the supraspinatus muscle as well as articular branches to both the glenohumeral and AC joints. The nerve then turns around the lateral edge of the base of the spine of

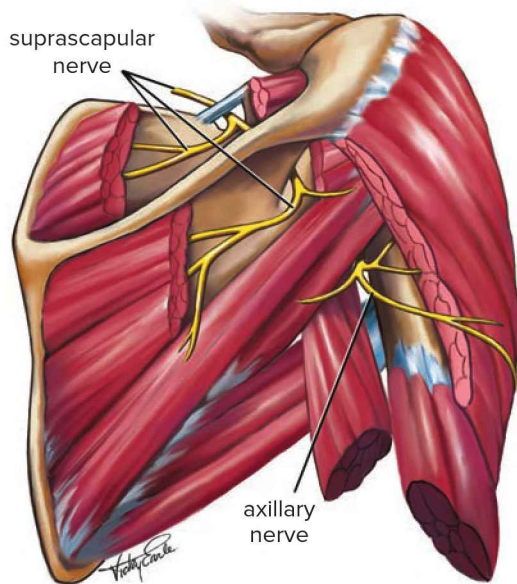


Figure 24.55 Course of the suprascapular nerve

the scapula (the spinoglenoid notch) to innervate the infraspinatus muscle. Entrapment of the suprascapular nerve may occur at either the suprascapular notch or the spinoglenoid notch.

The patient usually complains of pain that is deep and poorly localised. It is often felt posteriorly and laterally in the shoulder, or referred to the arm, neck or upper anterior chest wall. The patient may describe shoulder weakness. On examination, there may be wasting of the supraspinatus and/or infraspinatus muscles accompanied by weakness on abduction and external rotation. Tenderness over the suprascapular notch may also be present.

The site of entrapment in cases of combined supraspinatus and infraspinatus weakness is the suprascapular notch. The nerve may be stretched and kinked in this position by extremes of scapular motion associated with the throwing action. It may also occur in tennis players who complain of weakness and lack of control over backhand volleys. Diagnosis is made on the clinical symptoms and confirmed by an abnormal electromyogram. Surgical decompression of the nerve at the site of entrapment is occasionally required.

Isolated infraspinatus weakness and wasting may occur when the suprascapular nerve is trapped at the spinoglenoid notch. This condition has been seen in volleyball players who use the 'float' serve^{106, 107} and in weight lifters. It can also arise due to a cyst that results from superior glenoid labral tears compressing the nerve. Treatment should be directed to repairing the labral tear.

The main task of the coach and sports physician is to monitor external rotation strength on a regular basis (for instance using a hand-held dynamometer) in an athletic population known to be at risk for suprascapular nerve pathology such as volleyball players, and to give an external rotator strengthening program in the injury prevention program. In particular, focusing on external rotation strength training in several positions (neutral, abduction, forward flexion) guarantees sufficient training of all external rotators, infraspinatus as well as teres minor.¹⁰⁸

Long thoracic nerve injury

The long thoracic nerve is formed from the roots of the C5, C6 and C7 nerves. The nerve passes behind the brachial plexus to perforate the fascia of the proximal serratus anterior, passing medial to the coracoid with branches throughout the length of the serratus anterior. Long thoracic nerve palsy causes paralysis of the serratus anterior, with winging of the scapula. The nerve may be injured by traction on the neck or shoulder or by blunt trauma. Isolated long thoracic nerve palsy may also follow viral illnesses.

Clinical features include pain and limited shoulder elevation. Patients may complain of difficulty in lifting weights or an uncomfortable feeling of pressure from a chair against a winged scapula while sitting. They may also develop secondary impingement due to poor scapular control. The most striking feature on examination is winging of the scapula when pushing against a wall with both hands. Electromyographic studies will confirm the diagnosis. Initial treatment is conservative and most patients will recover fully. During the rehabilitation period, it is extremely important to give the patient exercises in which the winging is minimally present or suppressed. Very often, this is only possible with low load closed chain exercises. Surgical tendon transfer may occasionally be required.

Axillary nerve injury

Axillary nerve compression or quadrilateral space syndrome is an uncommon condition caused by compression of the posterior humeral circumflex artery and axillary nerve or one of its major branches in the quadrilateral space.¹⁰⁹ The quadrilateral or quadrangular space is located over the posterior scapula in the subdeltoid region and consists of the teres minor superiorly, teres major inferiorly, the long head of triceps medially and the surgical neck of the humerus laterally. The axillary nerve and the posterior humeral circumflex artery pass through the space at a level inferior to the glenohumeral joint capsule.

Quadrilateral space syndrome is seen in throwing athletes and is characterised by poorly localised posterior shoulder pain, paraesthesia over the lateral aspect of the shoulder and arm, and deltoid and teres minor weakness. The condition may occur secondary to abnormal fibrous

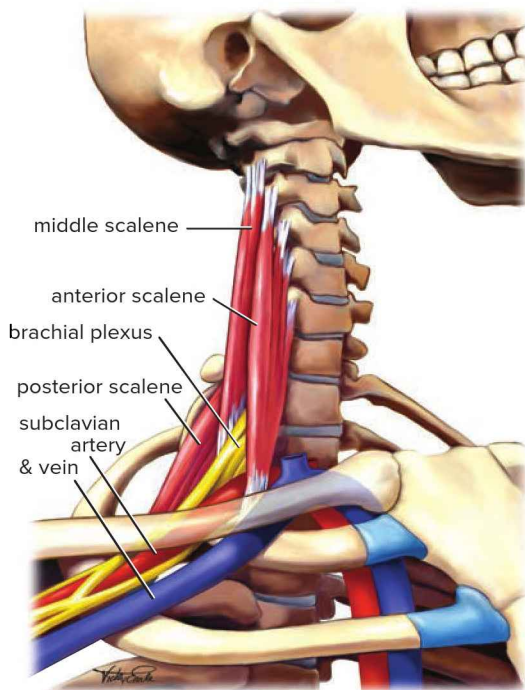


Figure 24.56 Anatomy of the thoracic outlet

bands, although traumatic causes have been described. Diagnosis is by electromyography or subclavian arteriogram, although this is associated with some risk. Treatment is initially conservative. Occasionally, surgical exploration is required.

Axillary nerve injuries can also occur with anterior dislocation of the shoulder and by blunt trauma to the anterior lateral deltoid muscle.¹¹⁰

Thoracic outlet syndrome

The term thoracic outlet syndrome (TOS) refers to a group of conditions that result from compression of the neurovascular structures that course from the neck to the axilla through the thoracic outlet (Fig. 24.56). The brachial plexus and subclavian vessels are especially susceptible to compression because of their proximity to one another in the thoracic outlet. The most common site of compression is the costoclavicular space between the clavicle and the first rib (costoclavicular syndrome).¹¹¹ Other sites of compression are the triangle between the anterior scalene muscle, the middle scalene muscle and the upper border of the first rib (anterior scalene syndrome); and the angle between the coracoid process and the pectoralis minor insertion (hyperabduction syndrome or pectoralis minor syndrome).

This condition may occur in overhead athletes. Poor posture with drooping shoulders and scapular protraction can decrease the diameter of the cervicoaxillary canal, causing TOS symptoms.

Congenital anatomical abnormalities including complete cervical rib, incomplete cervical ribs with fibrous bands, fibrous bands from the transverse process of C7, and clavicular abnormalities, can all compress the neurovascular structures. Complete cervical ribs are rare but, if present, are often bilateral. However, only 10% of patients with cervical ribs have TOS symptoms.

Traumatic structural changes that can cause TOS include fractures of the clavicle and/or first rib, pseudoarthrosis of the clavicle, mal-union of clavicle fractures, exuberant callus formation or a crush injury of the upper thorax. TOS symptoms are common in patients with chronic scapular dyskinesis. They have tight pectoralis minor, scalene and upper trapezius muscles, with weak serratus anterior and lower trapezius muscles; this causes excessive anterior tilt and protraction.

Clinical features

While patients with TOS occasionally present with a pure arterial, venous or neurogenic picture, most often the picture is mixed. The patient with TOS may complain of pain in the neck or shoulder, or numbness or tingling involving either the entire upper limb or the forearm and hand. The patient may state that the arm feels weak or easily fatigued. There may be venous engorgement or coolness of the involved arm.

Physical signs may be absent. A patient with arterial compression may have a positive Adson's test. The patient begins the test with the head laterally rotated to the side of the symptoms and extended. The patient then abducts the involved arm and inspires deeply. A positive test obliterates the radial pulse and reproduces symptoms. The sensitivity of this test can be greatly increased by the use of Doppler flow patterns during the manoeuvre. The most sensitive provocation test is the Roos hyperabduction/external rotation test (Fig 24.13a), in which the patient opens and closes his or her hands for 1–3 minutes with elbows bent and arms abducted to 90° and externally rotated in an attempt to reproduce the symptoms. Evaluation of scapular motion and position can help to rule scapular dyskinesis in or out.

Treatment

Treatment for any subset of TOS focuses on the specific area compromised. However, certain treatments apply to all forms of TOS. Correction of drooping shoulders, poor posture and poor body mechanics is vital. The patient should be taught proper positioning while sitting, standing and lying down. Physiotherapy should include pectoral and scalene stretching and trigger point treatment, and soft tissue mobilisation of restricted tissues, as well as scapular and scapulothoracic mobilisation. Joint mobilisation of the first rib can restore accessory motion of the sternoclavicular

and AC joints. Therapeutic exercise, education and manual therapy can correct forward head posture. Thoracic extension and brachial plexus neuromobility exercises are added as tolerated.

Surgical consultation and treatment is warranted for patients who have neurogenic TOS that does not respond to aggressive conservative management and for patients who have vascular compromise or thrombus formation.¹¹² Arterial compression caused by a complete cervical rib is usually treated by first rib resection.

Axillary vein thrombosis ('effort' thrombosis)

Axillary vein thrombosis is also known as 'effort' thrombosis because of its frequent association with repetitive, vigorous activities or with blunt trauma that results in direct or indirect injury to the vein. The axillary vein can be compressed at various sites along its path, most significantly in the costoclavicular space. Compression most often occurs when the patient hyperextends the neck and hyperabducts the arm simultaneously, or when the patient assumes a military brace position with a backward thrust of the shoulders. Compression can also occur between the clavicle and the first rib, the costocoracoid ligament and first rib, or the subclavian muscles and first rib.

Patients complain of dull, aching pain, numbness or tightness, and heaviness of the upper arm and shoulder, along with fatigue after activities involving the extremity. In axillary vein thrombosis, the entire upper extremity will be swollen, the skin may be mottled and cold, and superficial veins may be prominent. The diagnosis is confirmed on venography.

Treatment involves immediate anticoagulant therapy. Most patients make a full recovery and are able to resume sporting activities.¹¹³

Less common fractures around the shoulder

Stress fractures around the shoulder joint are uncommon. Stress fracture of the coracoid process is associated with the sport of trapshooting. Patients with this stress fracture have localised tenderness over the coracoid process and a focal area of abnormality on MRI or isotopic bone scan. Another specific site of stress fractures is anterolateral rib fractures in rowers, probably due to excessive activation of the serratus anterior (see Chapter 9).

Scapular fractures are usually due to a crushing force, either a fall on the shoulder or direct violence. Examination reveals marked tenderness and swelling. X-rays should be taken to exclude other associated injuries, such as a rib fracture, dislocated shoulder or dislocated sternoclavicular joint. Scapular fractures usually heal well, even if displaced. A broad arm sling is worn for comfort and active movements are commenced as soon as pain permits.

Fracture of the neck of the humerus is caused by a fall on the outstretched hand or direct violence.¹¹⁴ It is seen in adolescents, young adults and the elderly. Fractures involving more than two fragments displaced by more than 1 cm or associated with shoulder dislocation require surgical assessment. Minimally displaced or angulated fractures may be treated conservatively. Impacted fractures heal rapidly and can be supported in a broad arm sling. Displaced fractures are best treated in a collar and cuff that allows gravity to correct any angulation. For the first 2 weeks the arm should be kept in a sling under a shirt. After 2 weeks, pendular movement exercises of the shoulder joint should be commenced. From 4 weeks, a collar and cuff may be worn outside the clothes and gradually removed in stages over the following 2 weeks.

An unusual fracture is seen among throwing athletes. These athletes sustain a closed external rotation spiral fracture of their humerus immediately below the insertion of the deltoid muscle at the junction of the middle and lower thirds of the humerus or along the radial groove. Many, but not all, patients with this fracture give a history of pain at the site of the fracture so it may be regarded as a rapid-onset stress fracture. The fracture heals well in a cast or functional brace.

Snapping scapula

The scapula is surrounded by a number of scapular bursae to allow smooth movement over the thoracic wall. Snapping scapula syndrome or scapulothoracic bursitis is a condition characterised by periscapular crepitus and pain during arm movements. While some patients are mildly symptomatic, others may complain of severe pain and poor shoulder function even with simple tasks.

Snapping scapula is the result of a lack of congruence between the concave scapula and the convex thoracic wall, which can occur from anatomic predisposition, space-occupying skeletal lesions, fibrotic bursae, poor thoracic posture or scapular muscle control.¹¹⁵ Often the scapulothoracic bursitis is the result of combined structural (fibrotic bursae, elastofibroma) and functional (tightness of surrounding musculature, for instance pectoralis minor) deficits.

Regardless of the aetiology, current data support initial conservative treatment in patients with symptomatic snapping scapula. Non-operative treatment consists of non-steroidal anti-inflammatory medications, therapeutic injections of steroids and local anaesthetics into the inflamed bursa, activity modification and physiotherapy. Surgical treatment, bursectomy, is indicated after failure of conservative management. However, if anatomical lesions or deformities capable of producing scapular snapping are identified on imaging, primary surgical intervention is the favourable treatment option. Radiographic evaluation of the

snapping scapula is important to rule out osseous deformity, tumours or other bony causes of the patient's pain. Three-dimensional CT and MRI aid in detecting muscle, bony and soft tissue abnormalities.¹¹⁶

SPECIAL CONSIDERATIONS FOR THE OVERHEAD ATHLETE

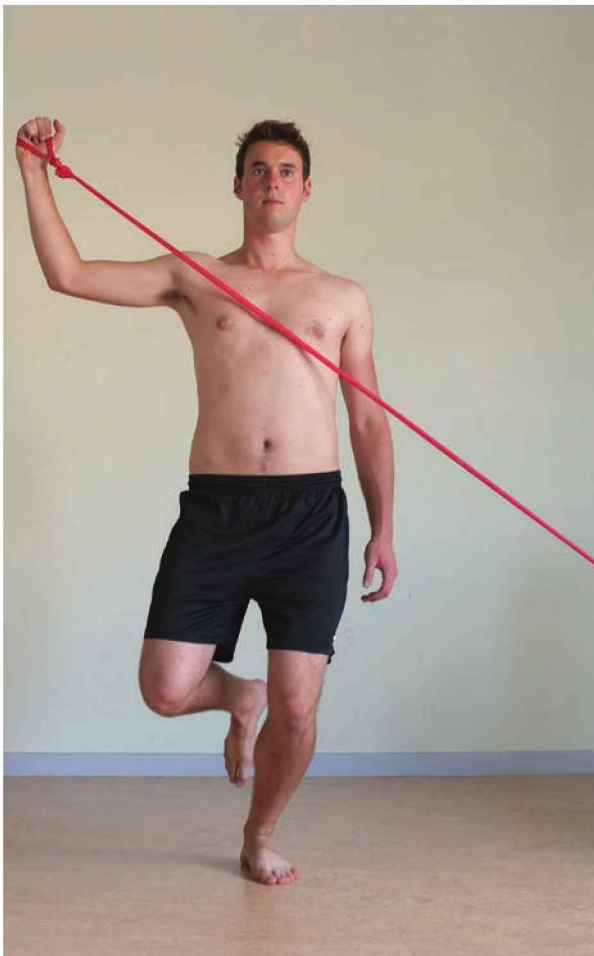
In any overhead athlete with a shoulder injury, the role of the kinetic chain (Chapter 8) must be considered.

Kinetic chain integration

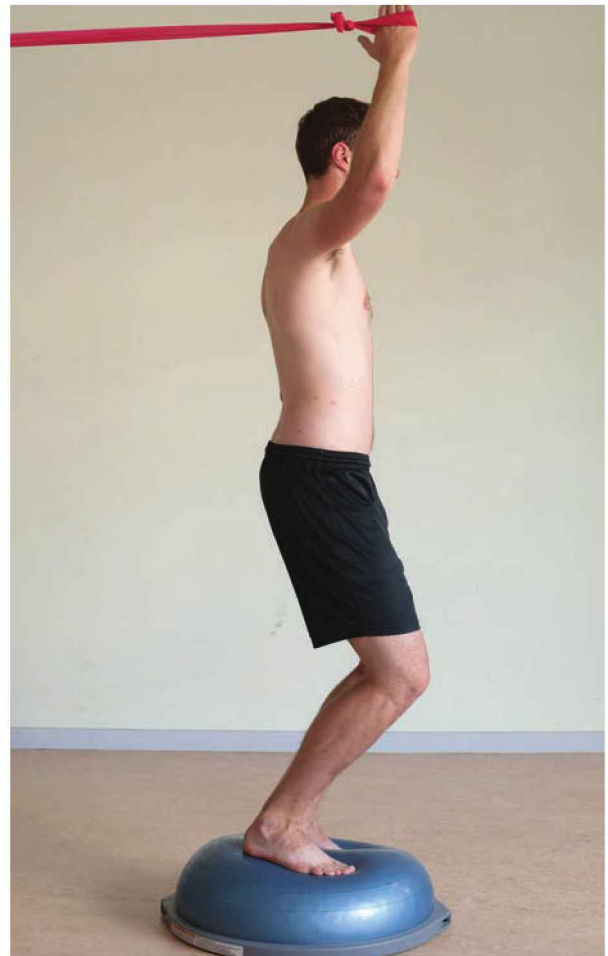
It is important to re-establish the kinetic chain early in the rehabilitation process. While the shoulder is recovering from the injury or surgery, leg and trunk exercises can be prescribed so that when the shoulder is ready for rehabilitation, the base of the kinetic chain is also ready for linked activity. After the shoulder is

ready for rehabilitation, sport-specific activation of the kinetic chain patterns restores the force-dependent motor activation patterns and normal biomechanical positions. This then allows normal link sequencing to generate velocity and force.

The athlete's kinetic chain is highly sport-specific, and varies for every athlete. In ground-based sports, like baseball, tennis or cricket, all of the activities of the shoulder work within a kinetic chain linkage from the ground through to the trunk, mostly in a diagonal pattern. These athletes benefit from diagonal pattern exercises in a closed chain for the lower extremities. In addition, performing shoulder exercises standing on the contralateral leg enhances scapular muscle activity¹¹⁷ (Fig. 24.57a). However, sports like volleyball, gymnastics and swimming have totally different movement patterns, and thus different demands from their kinetic chain. Volleyball players load their shoulder the most

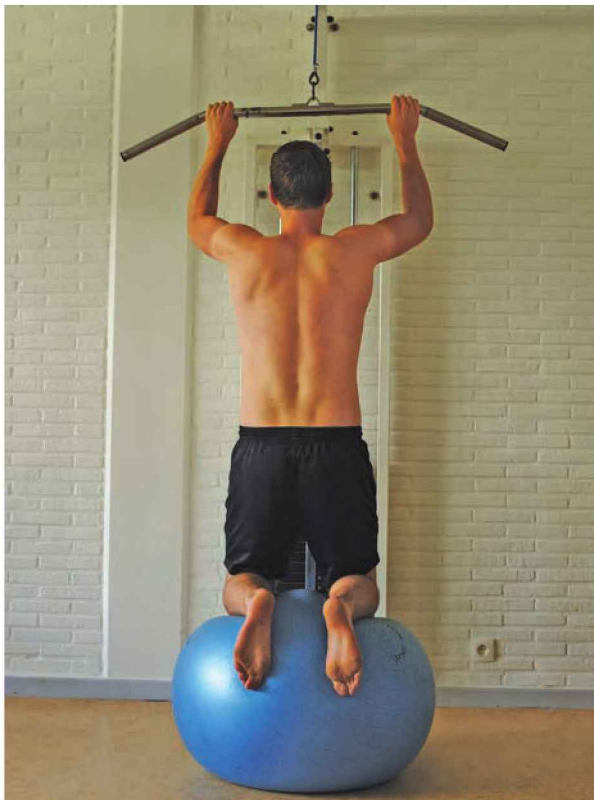


(a)



(b)

Figure 24.57 (a) Diagonal external rotation exercise standing on the contralateral leg (b) Throwing exercise on an unstable surface



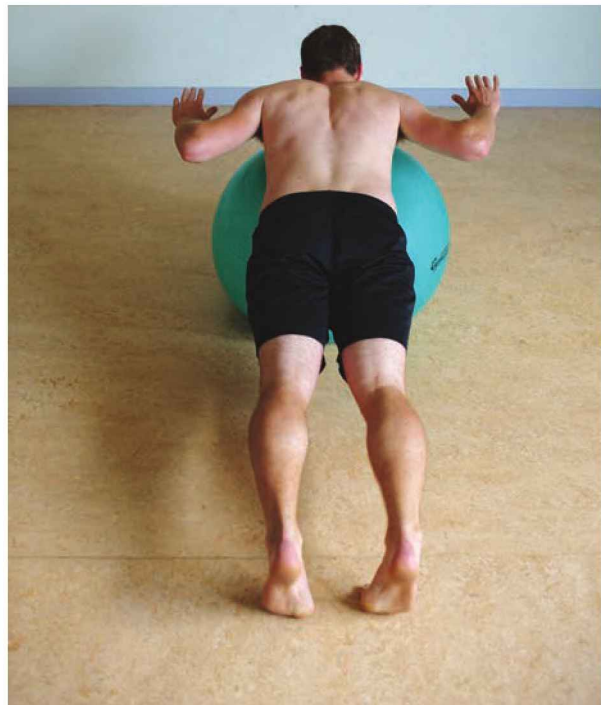
(c)

Figure 24.57 (cont.) (c) Progressive 'training hanging' for gymnasts

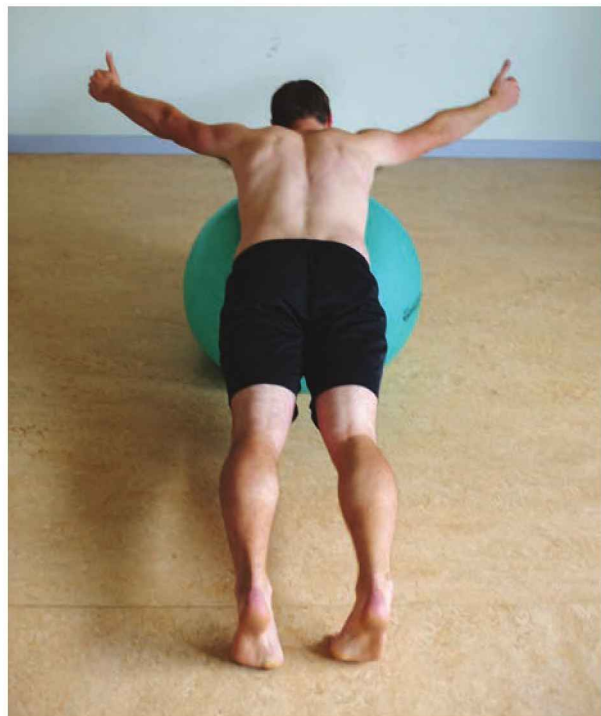
during smashing, when their feet are off the ground. They should train the smashing and decelerating capacity of their shoulders with minimal input from the ground, for instance on an unstable surface (Fig. 24.57b). Gymnasts should train in hanging (Fig. 24.57c) or high load closed chain positions mimicking their sport-specific positions. Swimmers should focus on endurance in a concentric manner in a prone (Fig. 24.58) and supine position. Tennis (for the two-handed backhand), hockey and golf players should additionally pay attention to movements using both hands. Athletes who have to be able to 'fall' (diving in volleyball, goal keepers in soccer) should be trained to do so, for instance using sliding boards (Fig. 24.59).

PRACTICE PEARL

It is important to correct any inflexibilities of the hamstrings, hip and trunk; weakness or imbalances of the rotators of the trunk, flexors and extensors of the trunk and hip; and any subclinical adaptations of stance patterns or gait pattern.



(a)



(b)

Figure 24.58 Prone exercise W-V—patient moves the arms from (a) Elbow in the back pockets (W) up to (b) High elevation (V)



Figure 24.59 Exercise training for impact on a gliding surface

In ground-based sports, rehabilitation of the legs and hips should be concerned with generating appropriate sport-specific force and velocity from the lower extremity and should be done in a closed chain. This pattern, which is done with the foot on the ground, simulates the patterns that exist in the throwing or hitting activities. Eccentric patterns should also be emphasised to absorb the load from jumping forward movement or stopping of the plant leg in the baseball throw. Combined patterns of hip and trunk rotation in both directions, hip and shoulder diagonal patterns should also be emphasised. Integration of the scapular retraction muscles to the hip is very important because these reactions tend to be coupled in the cocking phases

of throwing. Hip extension and trunk stability should be trained performing squatting exercises.

Endurance activities in the legs should also be emphasised, as should aerobic endurance for recovery from exercise bouts and anaerobic endurance for agility and power work. Examples include mini-trampoline exercises, agility drills with running and jumping, jumping jacks and slider or fitter boards.

When working in closed chain, the kinetic chain activity also benefits from diagonal patterns. Research has shown that lower trapezius activity is increased when, during the four-point kneeling position, the contralateral leg is extended.¹¹⁸

The thrower's program

Various thrower's programs are published, all aiming at progressively training the shoulder in order to prepare the athlete to throw in baseball, serve in tennis, smash in volleyball.⁷⁴ The main goal is to increase the challenge versus throwing by (a) changing shoulder position from neutral to full abduction/external rotation (ABER), (b) gradually increasing speed of movement, (c) implementing sport-specific kinetic chain skills, such as diagonals and core stability, into the shoulder exercises. For the latter we refer to the previous paragraph. All the following exercises should be considered to be functional, meaning adding kinetic chain principles where appropriate, depending on the specific sports; they are described below in view of shoulder loading only.

The goals and types of exercises are summarised in Table 24.7.

Table 24.7 Which exercises to prescribe depends on the clinician's goal

Goal	Type of exercise
Basic RC training	ER and IR in neutral
From neutral to functional ABER	ER and IR diagonal with limited load in ABER
Increasing load in ABER	'Throwing' movement slow with high load in ABER
Increasing speed in ABER—low load	Quick low-load ER and IR movements in ABER
Functional throwing and decelerating	'Throwing' movement fast with high load in ABER
Focus on eccentric strength of the external rotators	Slow high-resistance eccentric exercise (avoid or limit concentric phase to enhance eccentric loading)
Focus on deceleration of the posterior shoulder muscles	'Catching' exercises
Focus on stretch-shortening phase of throwing	Use material that challenges plyometric cycle of throwing

RC = rotator cuff; ER = external rotation; IR = internal rotation; ABER = abduction external rotation

The program starts with IR and ER exercises with the shoulder in neutral position (Fig. 24.21b and c) or in slight elevation in the scapular plane. These positions put no strain on the shoulder capsule and allow strengthening in a safe position; however, they should be considered as non-functional from a sport-specific point of view. ER may also be performed with the athlete side-lying, to enhance proper UT/LT muscle balance (Fig 24.45a). If allowed (depending on the injury), eccentric drills in neutral may be performed, with side-lying catching exercises (Fig. 24.60). This exercise can be performed in the early stage, for instance in labral injury or instability, however, in case of muscle or tendon strain or overload injury, it should be postponed to a later stage.

The next stage of exercises brings the athlete's shoulder from neutral position to higher elevation angles and towards the throwing position; however, with limited load in that position. Examples are IR and ER diagonals towards or starting at the contralateral hip, standing sideways to the point of resistance (Fig. 24.61a and b). In addition, side-lying forward flexion and prone horizontal abduction with external rotation may be performed (see scapular rehabilitation). Eccentric exercises for the infraspinatus/trapezius may be implemented performing

side-lying catching exercises with the shoulder in 90° of forward flexion.

- To increase the challenge of the shoulder towards throwing positions, the load in this position is increased by facing the point of resistance (for ER) (Fig. 24.62a), or standing backwards towards it (for IR) (Fig. 24.62b), and performing slow throwing movements.
- After these rather slow exercises, speed is implemented in two steps: first, quick movements are performed in the ABER (abduction-external rotation) position into



Figure 24.60 Side-lying plyometric catching exercise



(a)

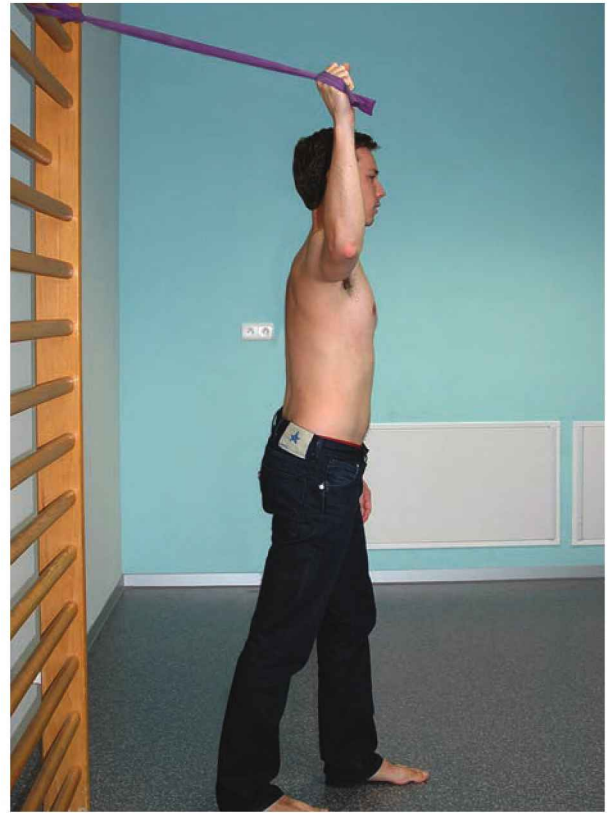


(b)

Figure 24.61 Diagonal (a) Internal and (b) External rotation with moderate load



(a)



(b)

Figure 24.62 The load is increased by facing the point of resistance (a) For ER or (b) Standing backwards towards it for IR

IR and ER without explosive strength, later on more explosive exercises, challenging the stretch-shortening cycle and plyometric nature of throwing are performed. During the first step, endurance is the exercise goal, and the athlete is encouraged to perform the exercise at two or more reps per second for 30 to 60 seconds consecutively. At the end of this stage, the athlete should also be able to perform catching exercises prone with the shoulder in ABER (Fig. 24.22).

- In the final stage of basic throwing exercises, the patient performs quick throwing movements against resistance to train the acceleration as well as the deceleration mechanism. In addition, the stretch-shortening cycle may be trained by the use of equipment that 'decelerates' the turning point between IR and ER (Fig. 24.23).
- In line with specific shoulder training, the therapist should not forget to pay additional attention to the elbow and the wrist and grip strength muscles.

Return to play following shoulder injury

According to the decision-based return-to-play model¹¹⁹ (Chapter 19), three steps need to be taken prior to full return to sports. In the first step, the health status of the athlete is evaluated, including assessment of symptoms and a battery of analytical and functional tests, for instance strength and flexibility, and throwing performance. Then the clinician evaluates the participation risk, based on the type of sport, level of competition and ability to protect the shoulder. Finally, some factors might modify the decision such as the timing in the season, pressure from the athlete or the environment. However, little evidence exists regarding the physical return-to-play criteria of the shoulder after injury. In particular, for each of the described risk factors, from a clinical perspective, there is a need for cut-off values to be used as criteria for return to train and return to play. In addition, the clinician needs objective and valid assessment tools applicable on the field or the

training area of the athlete, and finally, once deficits are assessed, there is a need for science-based training programs to restore normal values.

The following return-to-play criteria (Table 24.8), although not extensively supported by scientific evidence, may be used.^{120, 121}

Table 24.8 Return-to-play (RTP) criteria for the overhead throwing athlete following shoulder injury

Return-to-play criterion	Specific details
No/little pain	Self-report
Full ROM	Side difference GIRD <20°, TROM <5°
Normal strength	Dominant side 10% stronger in throwers, equal in strength in symmetric sports ER/IR ratio 66% (isokinetic), 75% (HHD in neutral), 95% (HHD in 90° abduction)
Normal scapular function	Slight scapular asymmetry allowed in resting position Scapular upward rotation +/- 60° in full elevation Dominant side 10% stronger in throwers, equal in strength in symmetric sports
Normal functional ability	Seated medicine ball throw Y-balance test Endurance test rotator cuff (elastic band, X-CO Trainer®)
Normal sport-specific skills	Having completed the thrower's program or another sport-specific program (non-throwing sports)

ROM = range of motion; GIRD = glenohumeral internal rotation deficit; TROM = total range of motion

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e